Notre Dame University Bangladesh

CSE Lab Reports

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Lab task Topic: Structured Programming Problems Solve & description

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Lab Report 01

1. Showing Hello World in output in using C language:

- **Program:**

```
//first program in c
#include <stdio.h>
//function main begins program execution
int main(void)
{
  printf("Hello, World!!");    //print the statement
   return 0;  //indicate that program ended successfully
}     //end function main
```

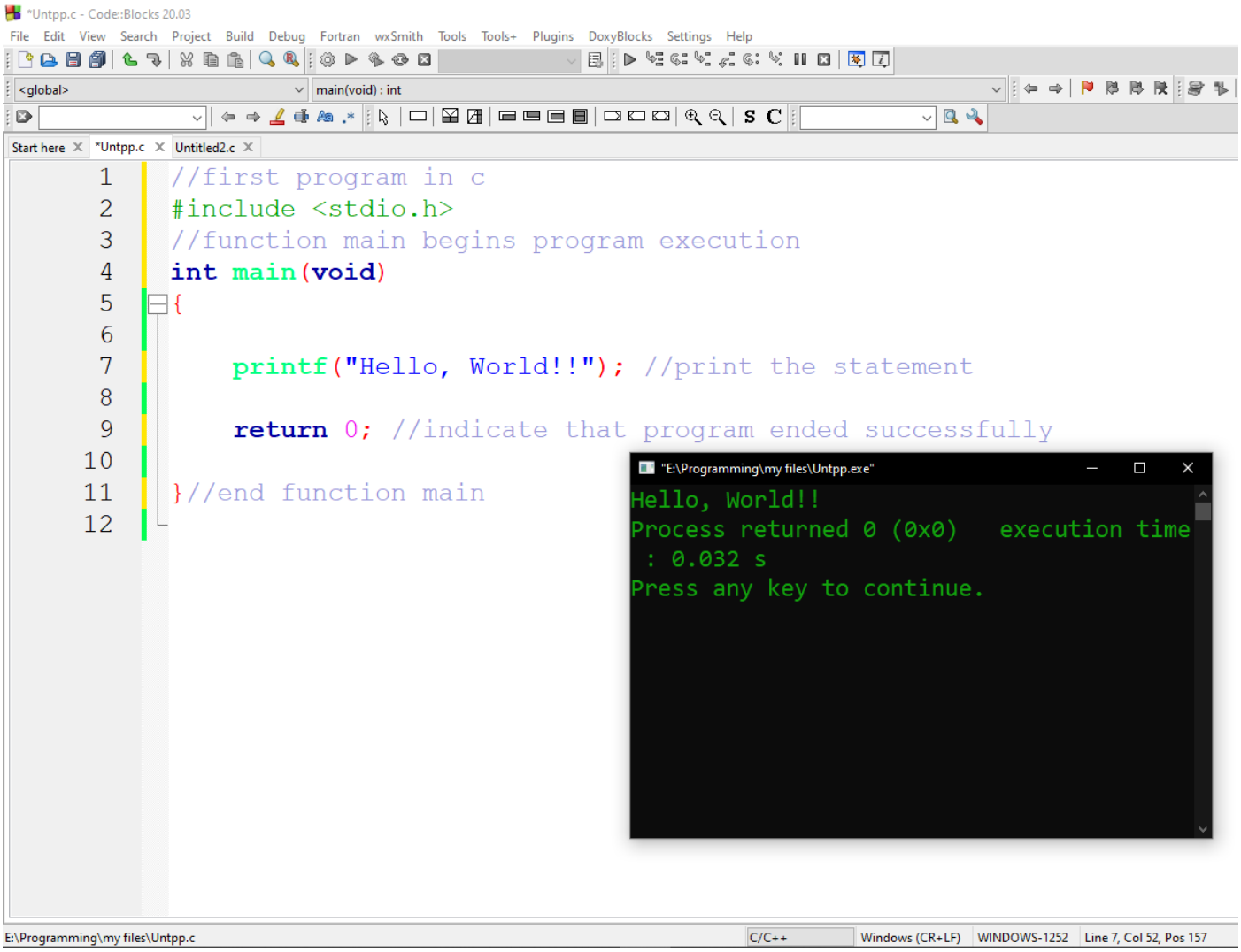


Figure 1.1 Program

Output:

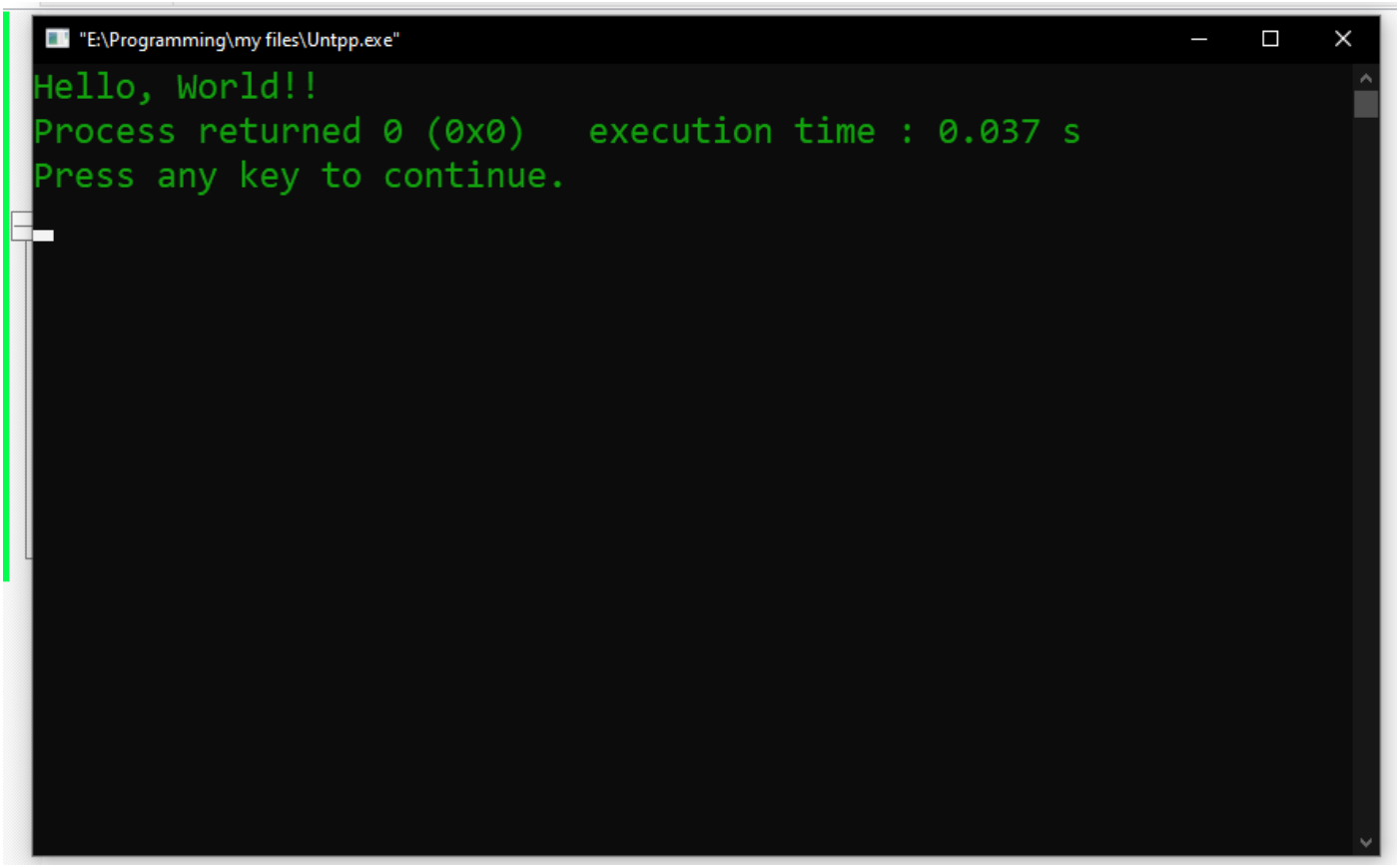


Figure 1.2 Output

- **Description:**

I begin by considering a simple C program in *Figure 1.1*

Begin with **//** indicating that these lines a comment, **"first program in C".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main (void)** all the function were written enter in the curly brace {}.

In line 7 I use **printf** for showing "Hello, world" in the monitor. It indicates that **printf** is supposed to do something out of the ordinary. The output is also here, as we can see the output in *Figure 1.2*

In line 9 I indicate that the program ended successfully by using **return 0;**

2. Summation of any 2 numbers using C language:

- **Program:**

```
//Summation of 2 numbers
#include <stdio.h>
//function main begins program execution
int main()
{
   int x, y, sum;                                  //assigning variables
   printf("Enter your first number:\n");          //user prompt
   scanf("%d", &x);                               //taking input from user
   printf("Enter your 2nd number:\n");            //user prompt
   scanf("%d", &y);                               //taking input from user
   sum=x+y; //summation of 2 numbers
   printf("The summation is:%d\n",sum);           //showing output
    return 0;                                     //indicate that program ended successfully
}                                                 //end function main
```

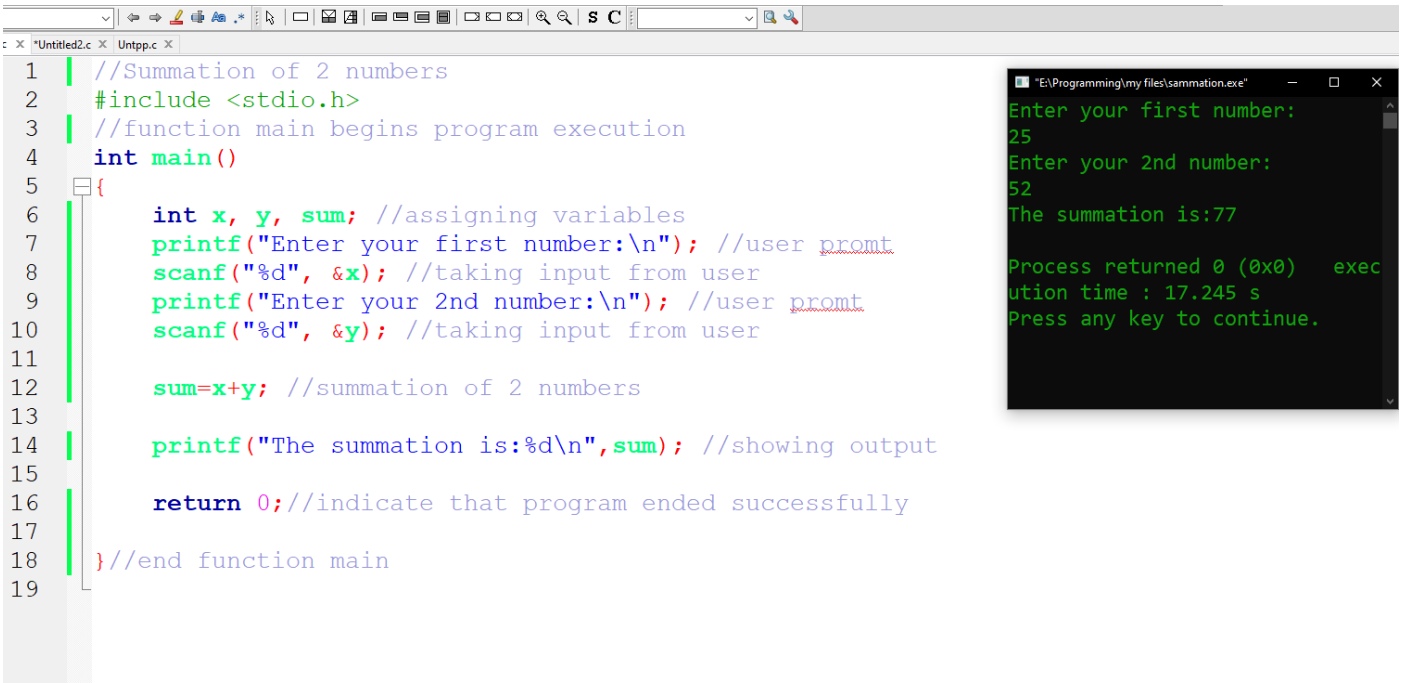


Figure 2.1 Program

Output:

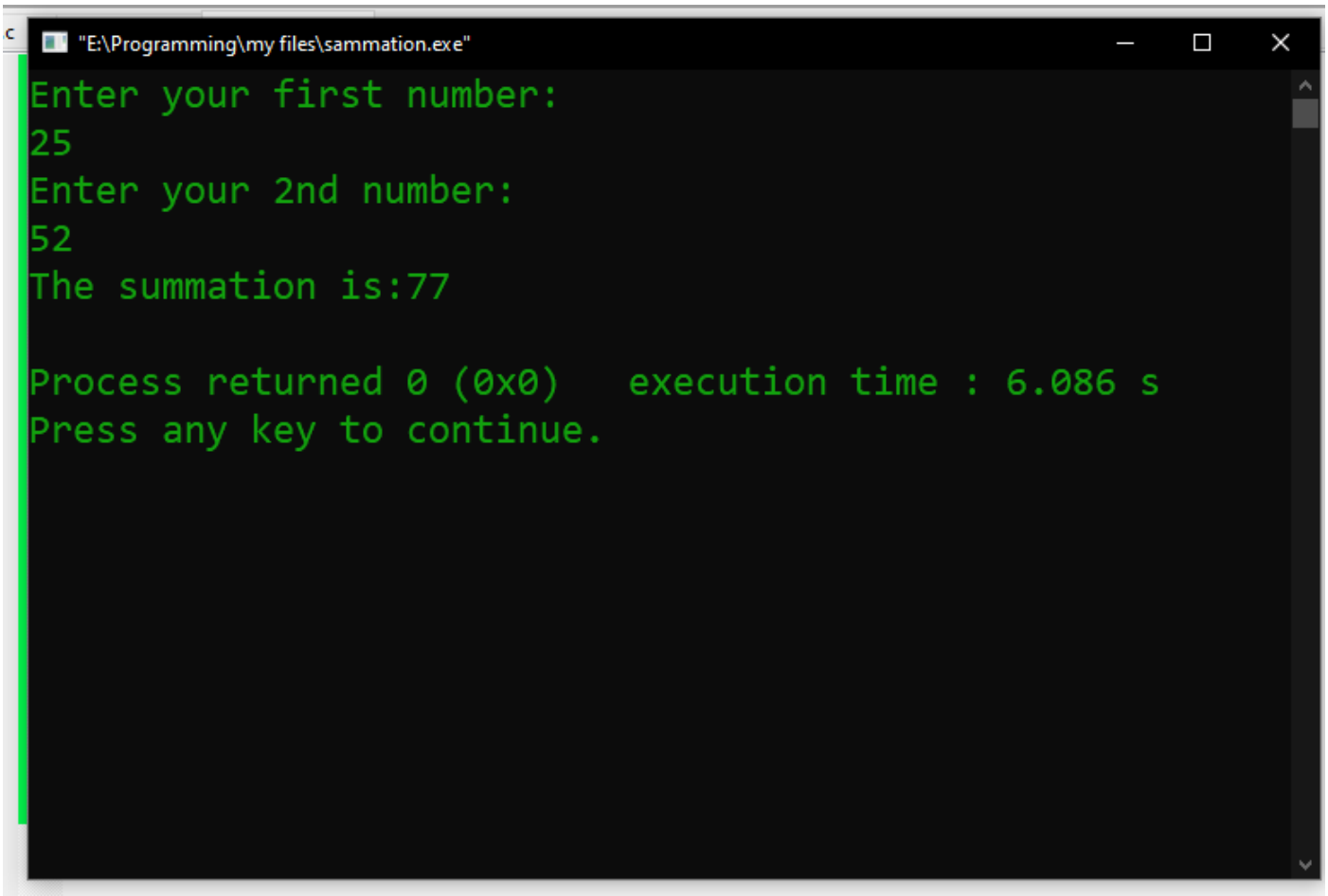


Figure 2.2 Output

- **Description:**

Begin with **//** indicating that this one line is a comment, **"Summation of 2 numbers".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main ()** all the function were written enter in the curly brace {}.

The names **x**, **y** and **sum** are the names of variables**.** In this program first of all I used **int x, y, sum;** This means that it needs to be worked on those variables. Then I used **printf** for user prompt, **"Enter the first number"**. Our next program uses the Standard Library function **scanf** to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using **printf** for showing "**The summation**" in the monitor. The output is also here, as we can see the output in *Figure 2.2*

In line 16 I indicate that the program ended successfully by using **return 0;**

3. Summation, Subtraction, Multiplication, Division of any 2 numbers using C language

- **Program:**

```
//summation, subtraction, Multiplication, Division of 2 numbers
#include <stdio.h>
//function main begins program execution
int main()
{
   int x, y, sum, sub, multi, div;                    //assigning variable
   printf("Enter your first number:\n");              //user promt
   scanf("%d", &x);//taking input from user
   printf("Enter your 2nd number:\n");               //user promt
   scanf("%d", &y);//taking input from user
   sum=x+y; //summation of 2 numbers
   sub=x-y; //subtraction of 2 numbers
   multi=x*y; //multiplication of 2 numbers
   div=x/y; //division of 2 numbers
   printf("The summation is:%d\n",sum);          //showing output
   printf("The subtraction is:%d\n",sub);
   printf("The multiplication is:%d\n",multi);
   printf("The division is:%d",div);
return 0; //indicate that program ended successfully
}//end function main
```

```c
//summation, subtraction, Multiplication, Division
#include <stdio.h>
//function main begins program execution
int main()
{
    int x, y, sum, sub, multi, div; //assigning variables
    printf("Enter your first number:\n");//user promt
    scanf("%d", &x);//taking input from user
    printf("Enter your 2nd number:\n");//user promt
    scanf("%d", &y);//taking input from user
    sum=x+y; //summation of 2 numbers
    sub=x-y; //subtraction of 2 numbers
    multi=x*y; //multiplication of 2 numbers
    div=x/y; //division of 2 numbers
    printf("The summation is:%d\n",sum); //showing output
    printf("The subtraction is:%d\n",sub);
    printf("The multiplication is:%d\n",multi);
    printf("The division is:%d",div);
return 0; //indicate that program ended successfully

}//end function main
```

Figure 3.1 Program

Output:

```
"E:\Programming\my files\Untitled2.exe"
Enter your first number:
12
Enter your 2nd number:
58
The summation is:70
The subtraction is:-46
The multiplication is:696
The division is:0
Process returned 0 (0x0)   execution time : 14.583 s
Press any key to continue.
```

Figure 3.2 Output

- **<u>Description:</u>**

Begin with **//** indicating that this one line is a comment, **"summation, subtraction, Multiplication, Division of 2 numbers".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main ()** all the function were written enter in the curly brace {}.

The names **x**, **y**, **sum, sub, multi, div** are the names of variables**.** In this program first of all I used **int x, y, sum, sub, multi, div;** This means that, it needs to be worked on those variables. Then I used **printf** for user prompt, **"Enter the first & second number"**. Our next program uses the Standard Library function **scanf** to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using **printf** for showing **the summation, subtraction, multiplication & division** one by one in the monitor. In the line 11 to 14 those Summation, subtraction, multiplication, division functions are written for processes the integer values of x & y variables. After processing the value, the output is also here, as we can see the output in *Figure 3.2*

In line 19 I indicate that the program ended successfully by using **return 0;**

Lab Report 02

Problem No. 01

Name of the problem:

Input two numbers from the keyboard, find out the summation of two numbers.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Summation of 2 numbers
#include <stdio.h>
//function main begins program execution
int main()
{
   int x, y, sum; //assigning variables
   printf("Enter your first number:\n"); //user prompt
   scanf("%d", &x); //taking input from user
   printf("Enter your 2nd number:\n"); //user prompt
   scanf("%d", &y); //taking input from user
   sum=x+y; //summation of 2 numbers
   printf("The summation is:%d\n",sum); //showing output
   return 0;//indicate that program ended successfully
}//end function main
```

Output:

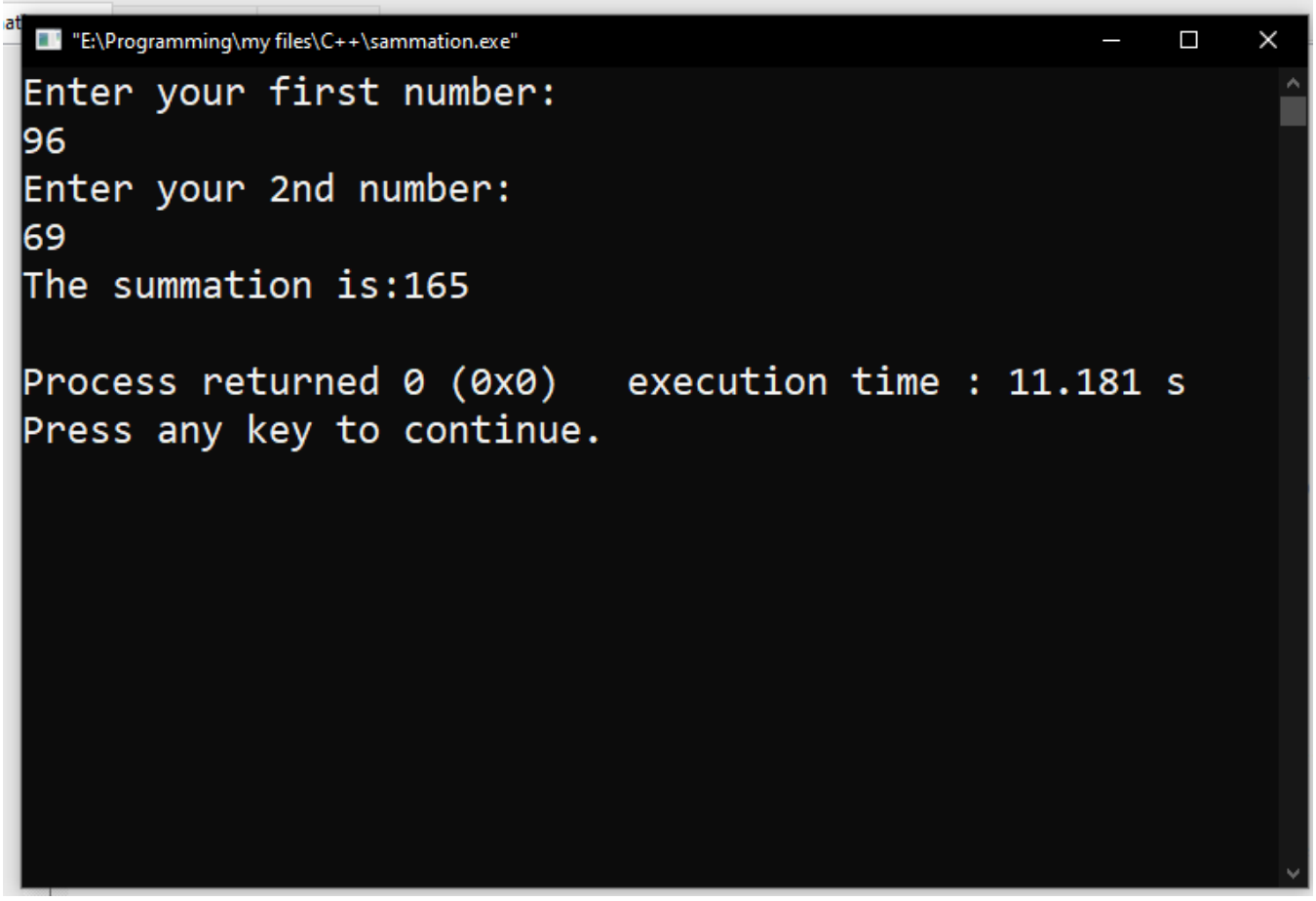


Figure 3 Output

Description:

Begin with // indicating that this one line is a comment, “Summation of 2 numbers”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}. The names x, y and sum are the names of variables. In this program first of all I used int x, y, sum; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the first number”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using printf for showing “The summation” in the monitor. The output is also here, as we can see the output in Figure 2.2 In line 16 I indicate that the program ended successfully by using return 0;

Problem No. 02

Name of the problem:

Write a program that requests two float type numbers from the user and then divides the first number by the second and display the result along with the numbers.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//division of 2 floating numbers

#include <stdio.h>

float main() //main function

{

   float x, y, div; //assign 2 integers

   printf("Enter your 2 numbers:\n"); //print user statement

   scanf("%f %f", &x, &y); //taking user input

   div=x/y;

   printf("Your division is: %f\n", div);

   return 0;

}
```

Output:

```
Enter your 2 numbers:
15.5
3.5
Your division is: 4.428571

Process returned 27 (0x1B)   execution time :
13.393 s
Press any key to continue.
```

Figure 4 Output

Description:

Begin with // indicating that this one line is a comment, “division of 2 floating numbers”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 3rd line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}. The names x, y and div are the names of variables.

In this program first of all I used int x, y, div; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the first number”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using printf for showing “The division” in the monitor. The output is also here, as we can see the output in Figure 2 In line 10 I indicate that the program ended successfully by using return 0;

Problem No. 03

Name of the problem:

Find out the area of a Rectangle shaped Room

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Area of a rectangle shaped room

#include <stdio.h>

int main()

{

   int l,w,area;

   printf("Enter the length value of the room :\n");

   scanf("%d", &l);

   printf("Enter the width value of the room:\n");

   scanf("%d", &w);

   area=l*w;

   printf("The area of the rectangle shaped room is:%d", area);

   return 0;

}
```

Output:

```
"E:\Programming\my files\C++\Area of ractangle.exe"
Enter the length value of the room :
50
Enter the width value of the room:
30
The area of the rectangle shaped room is:1500
Process returned 0 (0x0)   execution time : 6.
452 s
Press any key to continue.
```

Figure 5 Output

Description:

Begin with // indicating that this one line is a comment, “Area of a rectangle shaped room”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 3rd line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}. The names l, w and area are the names of variables.

In this program first of all I used int l, w, are; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the length value of the room” & “Enter the width value of the room ”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using area=l*w and prints the result using printf for showing “The area of the rectangle shaped room” in the monitor. The output is also here, as we can see the output in Figure 3 . Then I indicate that the program ended successfully by using return 0;

Problem No. 04

Name of the problem:

Find out the area of a Square shaped Room

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Area of a square Shaped room

#include <stdio.h>

int main()

{

   int l,area;

   printf("Enter the edge of the room :\n");

   scanf("%d", &l);

   area=l*l;

   printf("The area of the square shaped room is:%d", area);

   return 0;

}
```

Output:

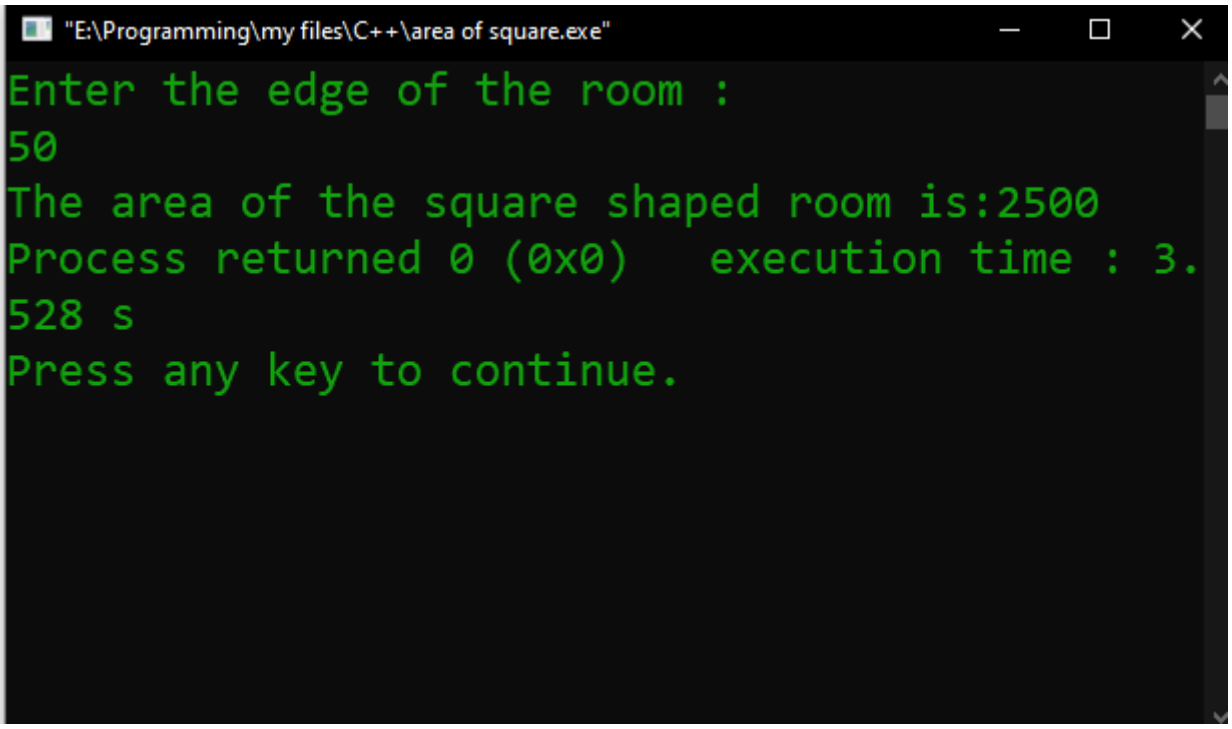


Figure 6 Output

Description:

Begin with // indicating that this one line is a comment, “Area of a square Shaped room”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names **l** and **area** are the names of variables. In this program first of all I used int l,area; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the edge of the room”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using area=l*l and prints the result using printf for showing “The Area of a square Shaped room” in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 05

Name of the problem

Find out the area of a triangle

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//area of a triangle
#include <stdio.h>
int main()
{
   int h,e,area;
   printf("Enter the height value of the triangle :\n");
   scanf("%d", &h);
   printf("Enter the edge value of the triangle:\n");
   scanf("%d", &e);
   area=0.5*(h*e);
   printf("The area of the triangle is:%d", area);
  return 0;
}
```

Output:

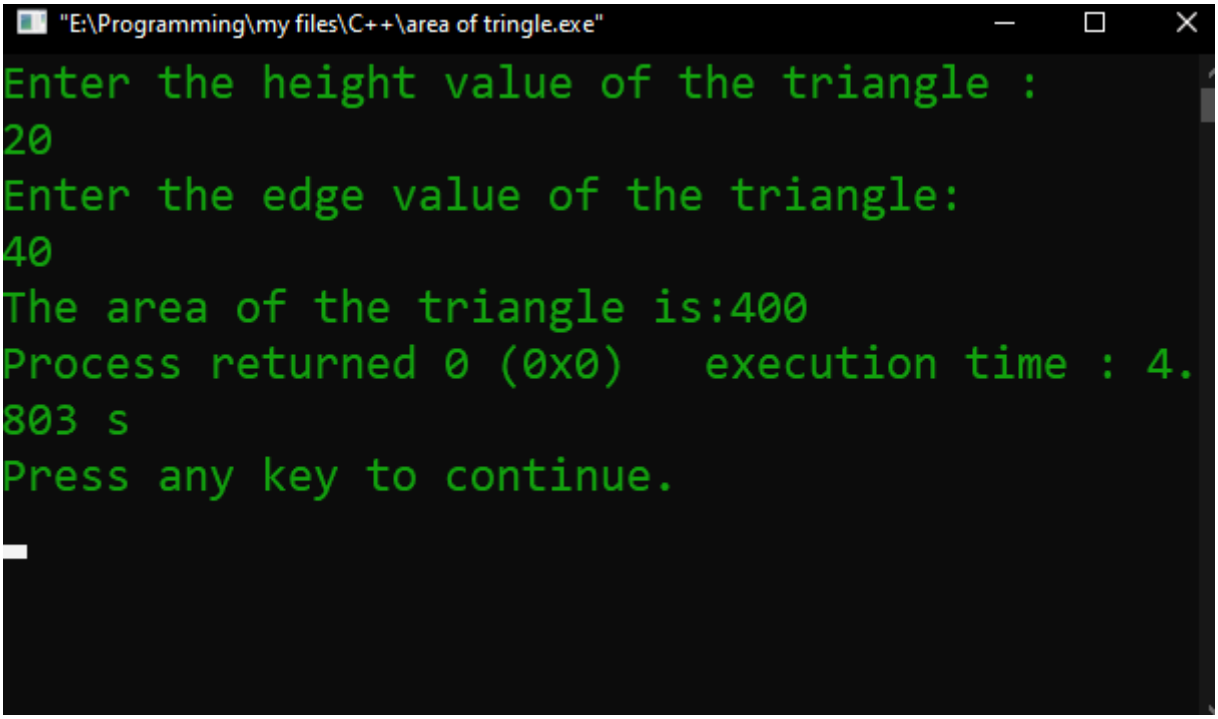


Figure 7 Output

Description:

Begin with // indicating that this one line is a comment, “Area of a triangle”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names **h, e** and **area** are the names of variables. In this program first of all I used int h,e,area; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the height value of the triangle” & ”Enter the edge value of the triangle”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using area=0.5*(h*e); and prints the result using printf for showing “The area of the triangle is:%d” in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Write down the programs 3, 4 and 5 in a single Code.

Program:

```
#include <stdio.h>
int main()
{
   int x,y,area,square,triangle;
   printf("Enter the length or height value :\n");
   scanf("%d", &x);
   printf("Enter the width value of the room:\n");
   scanf("%d", &y);
  area=x*y;
   printf("The area of the rectangle shaped room is:%d\n", area);
   square=x*x;
   printf("The area of the square is:%d\n", square);
   triangle=0.5*(x*y);
   printf("The area of the triangle is:%d\n", triangle);
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\rect, squ, tri.exe"
Enter the length or height value :
5
Enter the width value of the room:
4
The area of the rectangle shaped room is:20
The area of the square is:25
The area of the triangle is:10

Process returned 0 (0x0)   execution time : 7.
944 s
Press any key to continue.
```

Figure Output

Description:

Begin with // indicating that this one line is a comment, “area of the square,rectangle,”. I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names x,y,area,square & triangle are the names of variables. In this program first of all I used int x,y,area,square,triangle; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the height value” & ”Enter the edge value”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using and prints the result using printf for showing in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 06

Write a program to display the equation of a line in the form ‘ax+by=c’, for a=5, b=8 and c=18.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program;

```
//problem 6

#include <stdio.h>

int main()

{

  int a,b,c;

  printf("Enter the value of 'a':\n");

  scanf("%d",&a);

  printf("Enter the value of 'b':\n");

  scanf("%d",&b);

  c=(a*2)+(b*1);

  //printf("If 'ax+by=c'\n then, C=%d",C);

  printf("%dx+%dy=%d",a,b,c);

}
```

Output:

```
"E:\Programming\my files\C++\equations.exe"
Enter the value of 'a':
5
Enter the value of 'b':
8
5x+8y=18
Process returned 0 (0x0)   execution time : 3.
679 s
Press any key to continue.
```

Output

Description:

I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names int a,b,c are the names of variables. In this program first of all I used int a,b,c; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the value of 'a'” & ” Enter the value of 'b'”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using and prints the result using printf for showing in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem no. 07

Write a program to Converting temperature from Fahrenheit to Celsius and from Celsius to Fahrenheit

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//problem 7
#include <stdio.h>
int main()
{
   int x;
   printf("Converter!! \n press 1 for convert Celsius to Fahrenheit \n press 2 for convert
Fahrenheit to Celsius \n");
   scanf("%d", &x);
   if(x==1)
   {
      float c, F;
      printf("Enter the Celsius value:\n");
      scanf("%f",&c);
      F=(1.8*c)+32;
      printf("The Fahrenheit value is:%f",F);
   }
   else if(x==2)
   {
      float f,C;
      printf("Enter the Fahrenheit value:\n");
      scanf("%f",&f);
      C=((f-32)*5)/9;
      printf("The Celsius value is:%f",C);
   }
   else
      printf("your value is not valid");
}
```

Output:

```
"E:\Programming\my files\C++\c to f.exe"
Converter!!
 press 1 for convert Celsius to Fahrenheit
 press 2 for convert Fahrenheit to Celsius
1
Enter the Celsius value:
60
The Fahrenheit value is:140.000000
Process returned 0 (0x0)   execution time : 11.427
s
Press any key to continue.
```

Celsius to Fahrenheit

```
"E:\Programming\my files\C++\c to f.exe"
Converter!!
 press 1 for convert Celsius to Fahrenheit
 press 2 for convert Fahrenheit to Celsius
2
Enter the Fahrenheit value:
140
The Celsius value is:60.000000
Process returned 0 (0x0)   execution time : 16
.574 s
Press any key to continue.
```

Fahrenheit to Celsius

Problem No. 08

Suppose X=20, Y=10. Show the output as X has the value of Y and Y has the value of X.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include <stdio.h>

int main() //main function

{

   int  x, y;

   printf("Enter the value of X:\n");

   scanf("%d",&x);

  printf("Enter the value of:Y\n");

   scanf("%d",&y);

   x=x+y;
   y=x-y;
   x=x-y;

   printf("your value of x=%d\n",x);

   printf("your value of y=%d",y);

   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\Untitled.exe"
Enter the value of X:
20
Enter the value of:Y
10
your value of x=10
your value of y=20
Process returned 0 (0x0)   execution time : 11
.177 s
Press any key to continue.
```

Output of x y

Description:

I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace { }.

The names int x, y; are the names of variables. In this program first of all I used int x,y; This means that it needs to be worked on those variables. Then I used printf for user prompt, “Enter the value of X” &”Enter the value of Y”. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using and prints the result using printf for showing in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 09

There were 4687 people in a village last year. in this year, 349 people moved in and 280 people moved out of the village. How many people are there in this year?
After executing this code you reproduce this code for any number of people.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include <stdio.h>

int main()

{

    int x, y, z, p;

    printf("Enter number of people in the village last year:");

    scanf("%d", &x);

    printf("Enter the number of people moving in:");

    scanf("%d", &y);

    printf("Enter the number of people moving out:");

    scanf("%d", &z);

    p = (x + y) - z;

    printf("Number of people this year is : %d", p);

    return 0;

}
```

Output:

```
"E:\Programming\my files\C++\village peple.exe"
Enter number of people in the village last year:4687
Enter the number of people moving in:349
Enter the number of people moving out:280
Number of people this year is : 4756
Process returned 0 (0x0)   execution time : 44.495 s
Press any key to continue.
```

Figure 9 Output

Description:

I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace { }.

The name int x, y, z, p; are the names of variables. In this program first of all I used int x, y, z, p; This means that it needs to be worked on those variables. Then I used printf for user prompt. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using and prints the result using printf for showing in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 10

A butcher sells beef at 700 Taka per KG and mutton at 800 taka per kg. If we buy 3 kg of beef and 2 kg of mutton, and pay 4000 Taka, how much will be the change? After executing this code you reproduce this code in generalized form.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include <stdio.h>
int main()
{
    float paid, beef, mutton, change;

    printf("Amount being paid: ");
    scanf("%f", &paid);
    printf("Kilos of beef getting bought: ");
    scanf("%f", &beef);
    printf("Kilos of mutton getting bought: ");
    scanf("%f", &mutton);

    change = paid - (beef * 700 + mutton * 800);

    printf("Amount of change is : %f", change);

    return 0;
}
```

Output:

```
"E:\Programming\my files\C++\kosai kasem.exe"
Amount being paid: 4000
Kilos of beef getting bought: 3
Kilos of mutton getting bought: 2
Amount of change is : 300.000000
Process returned 0 (0x0)   execution time : 36
.273 s
Press any key to continue.
```

Figure 10 Output

Description:

I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace { }.

The name float paid, beef, mutton, change; are the names of variables. In this program first of all I used float paid, beef, mutton, change; This means that it needs to be worked on those variables. Then I used printf for user prompt. Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the calculate of these values by using and prints the result using printf for showing in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Lab Report 03

Problem No. 01

Name of the problem:

Write a C program to find the maximum in the given two numbers using the conditional operator

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//find the maximum number among two numbers
#include <stdio.h>
int main()
{
   int x,y;
   printf("Enter your first number:");
   scanf("%d",&x);
   printf("Enter your second number:");
   scanf("%d",&y);

   (x>y) ? printf("The largest number is:%d",x) : printf("The largest number
is:%d",x);
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\conditional 1.exe"
Enter your first number:52
Enter your second number:54
The largest number is:52
Process returned 0 (0x0)   execution time : 12
.104 s
Press any key to continue.
```

Figure 8 Output

Description:

Begin with // indicating that this one line is a comment, "//find the maximum number among two numbers". I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names x and y are the names of variables. In this program first of all I used int x, y; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the first number". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the compare of these values by using conditional operator and prints the result using printf for showing "The largest number" in the monitor.

The output is also here, as we can see the output in fig 1 In line 13 I indicate that the program ended successfully by using return 0;

Problem No. 02

Name of the problem:

Write a C program to find the inputted number is positive or negative using the conditional operator.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Find the inputted number is positive or negative

#include <stdio.h>


int main()

{

   int x;

   printf("Enter your first number:");

   scanf("%d",&x);


         (x>0) ? printf("The number is positive +%d",x) : printf("The number is negative
%d",x);

   return 0;

}
```

Output:

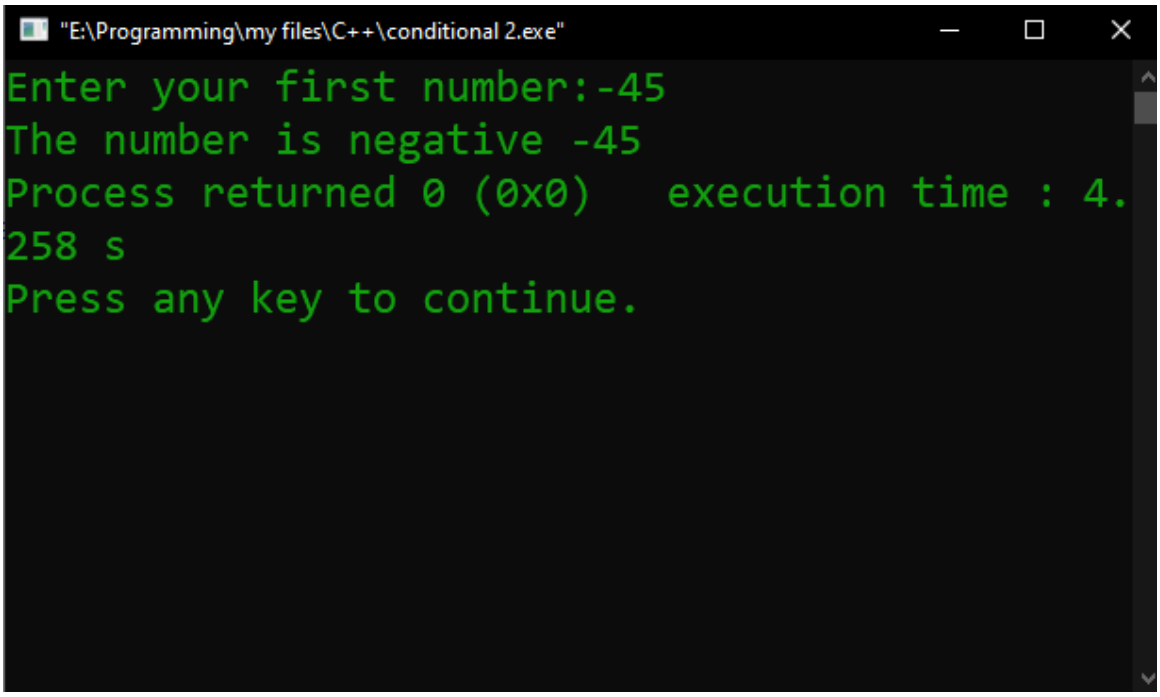


Figure 9 Output

Description:

Begin with // indicating that this one line is a comment, //Find the inputted number is positive or negative". I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names x is the names of variables. In this program first of all I used int x; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the first number". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the compare of these values by using conditional operator and prints the result using printf for showing "The positive or negative number" in the monitor.

The output is also here, as we can see the output in fig 2 In line 9 I indicate that the program ended successfully by using return 0;

Problem No. 03

Name of the problem:

Write a C program to check whether the given number is odd or even using the conditional operator in C

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//check whether the given number is odd or even

#include <stdio.h>


int main()

{

   int x;

   printf("Enter your first number:");

   scanf("%d",&x);


   (x%2==0) ? printf("The number is even: %d",x) : printf("The number is odd: %d",x);

   return 0;

}
```

Output:

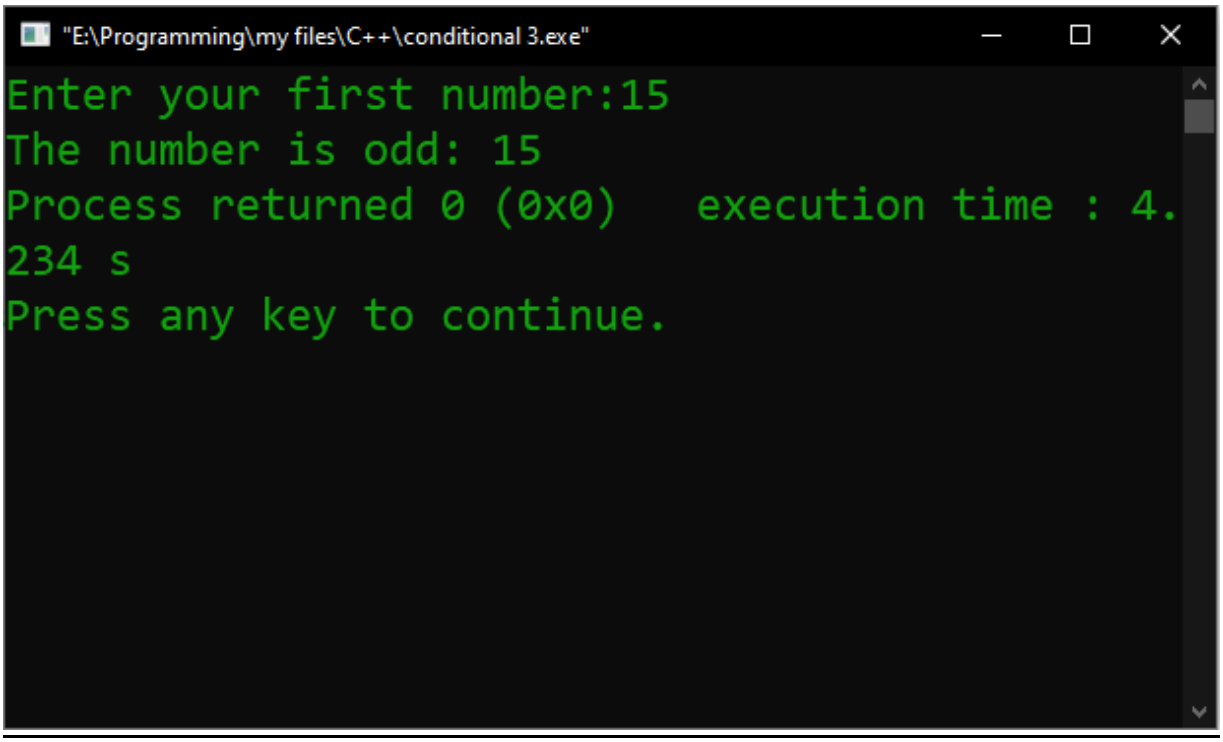


Figure 10 Output

Description:

Begin with // indicating that this one line is a comment, //check whether the given number is odd or even". I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names x is the names of variables. In this program first of all I used int x; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the first number". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the compare of these values by using conditional operator and prints the result using printf for showing "The number is odd or even" in the monitor.

The output is also here, as we can see the output in fig 3. In line 9, I indicate that the program ended successfully by using return 0;

Problem No. 04

Name of the problem:

Write a program to enter two numbers. Make a comparison between them with the conditional operator. If the first number is greater than the second, perform a division operation otherwise multiplication operation.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
/*Making a comparison between 2 numbers with the conditional operator. If the first
number is greater than the second, perform a division operation otherwise multiplication
operation*/

#include <stdio.h>

int main()

{

   int x,y,div,multi;

   printf("Enter your first number:");

   scanf("%d",&x);

   printf("Enter your second number:");

   scanf("%d",&y);

   (x>y) ? printf("the division of 2 numbers:%d",div=x/y) : printf("the multiplication of 2
numbers:%d",multi=x*y);

   return 0;

}
```

Output:

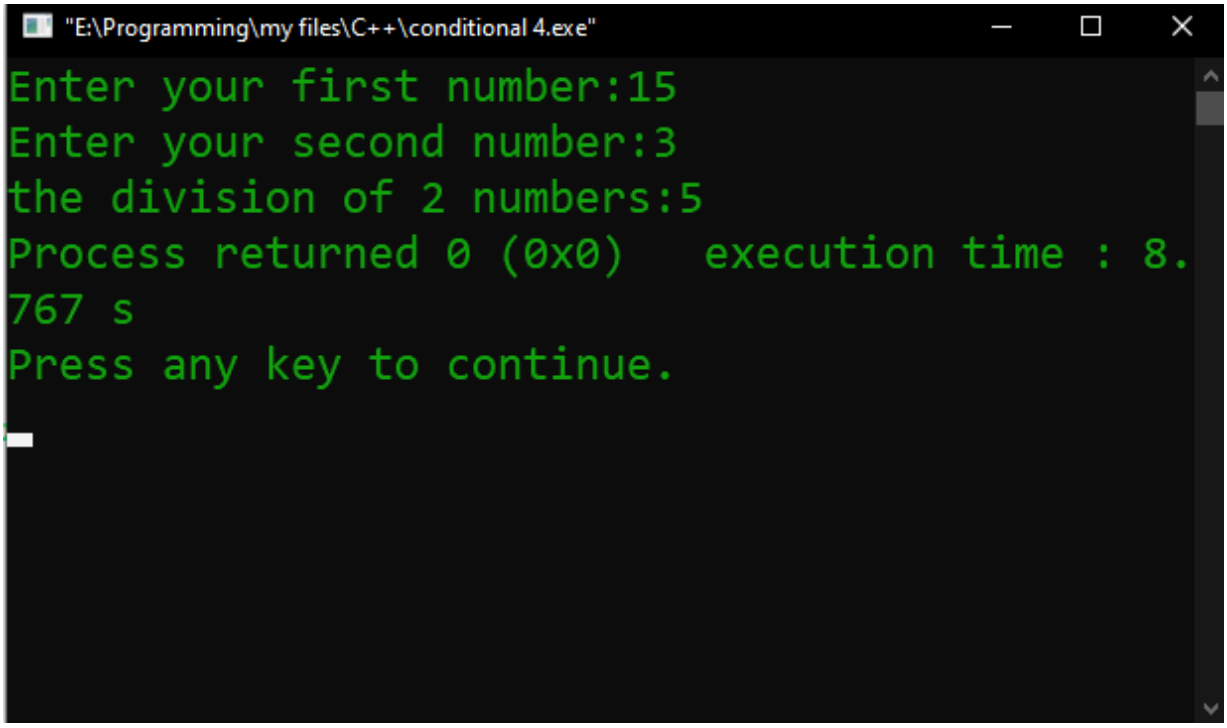


Figure 11 Output

Description:

Begin with // indicating that this one line is a comment, "/*Making a comparison between 2 numbers with the conditional operator. If the first number is greater than the second, perform a division operation otherwise multiplication operation*/". I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names **x, y, div** and **multi** are the names of variables. In this program first of all I used int x, y, div, multi; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the first number". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, compares & the calculate of these values by using div=x/y & multi=x*y and prints the result using printf for showing "The Division/ Multiplication of 2 numbers" in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 05

Name of the problem

Use the conditional operator to produce a code to have the following sample input and output. Sample Output for the different test-cases:-

Enter the number of items: 1 ----------→ You have 1 item.

Enter the number of items: 2 ----------→ You have 2 items.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include <stdio.h>


int main()
{
   int item;
   printf("Enter the number of items:");
   scanf("%d",&item);

   printf("you have %d item%s",item,(item==1) ? "" : "s");
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\conditional 5.exe"
Enter your first number:5
you have 5 items
Process returned 0 (0x0)   execution time : 6.
727 s
Press any key to continue.
```

Figure 12 Output

```
"E:\Programming\my files\C++\conditional 5.exe"
Enter your first number:1
you have 1 item
Process returned 0 (0x0)   execution time : 3.
048 s
Press any key to continue.
```

Figure 13 output

Description:

I use the conditional operator to produce a code to have the following sample input and output. Sample Output for the my test-cases:-

Enter the number of items: 5 ----------→ You have 5 items.

Enter the number of items: 1 ----------→ You have 1 item.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The name **item** is the names of variables. In this program first of all I used int item; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the number of items". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, compares & the calculate of these values by using Conditional Operator and prints the result using printf for showing "You have 5 items or You have 1 item" in the monitor. The output is also here, as we can see the output. I indicate that the program ended successfully by using return 0;

Problem No. 06

Find the largest among three numbers using the conditional operator.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program;

```
//Find the largest among three numbers using the conditional operator

#include <stdio.h>

int main()

{

   int x,y,z;

   printf("Enter your 3 numbers:\n");

   scanf("%d %d %d",&x,&y,&z);


   (x>y) ?

      ((x>z) ?

      printf("The largest number is %d",x):printf("The largest number is %d",z)):

      ((y>z) ?

      printf("The largest number is %d",y):printf("The largest number is %d",z));

   return 0;

}
```

Output:

```
"E:\Programming\my files\C++\conditional 6.exe"
Enter your 3 numbers:
100
500
200
The largest number is 500
Process returned 0 (0x0)   execution time : 12
.485 s
Press any key to continue.
```

Output 6

Description:

Begin with // indicating that this one line is a comment, "//Find the largest among three numbers using the conditional operator". I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run. In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called #include which is a directive to the C preprocessor. In the 4th line the main function begins execute the program. In the main function int main () all the function were written enter in the curly brace {}.

The names x, y, z are the names of variables. In this program first of all I used int x, y, z; This means that it needs to be worked on those variables. Then I used printf for user prompt, "Enter the 3 numbers". Our next program uses the Standard Library function scanf to obtain two integers typed by a user at the keyboard, computes the compare of these values by using conditional operator and prints the result using printf for showing "The largest number" in the monitor.

The output is also here, as we can see the output in fig 6 In line 13 I indicate that the program ended successfully by using return 0;

Problem no. 07

Write a C program to input a character and check whether the character is alphabet or not using Conditional operator.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//character check whether the character is alphabet or not
#include <stdio.h>
int main()
{
   char x;
   printf("Enter your character:");
   scanf("%c",&x);

   (x>='a' && x<='z') || (x>='A' && x<='Z') ?
   printf("The character is a:%c",x) : printf("The character is not an alphabet:%c",x);
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\condtional 7.exe"
Enter your character:A
The character is a:A
Process returned 0 (0x0)   execution time : 4.
592 s
Press any key to continue.
```

Figure 14

Problem No. 08

Write a C program to check leap year using Conditional operator.

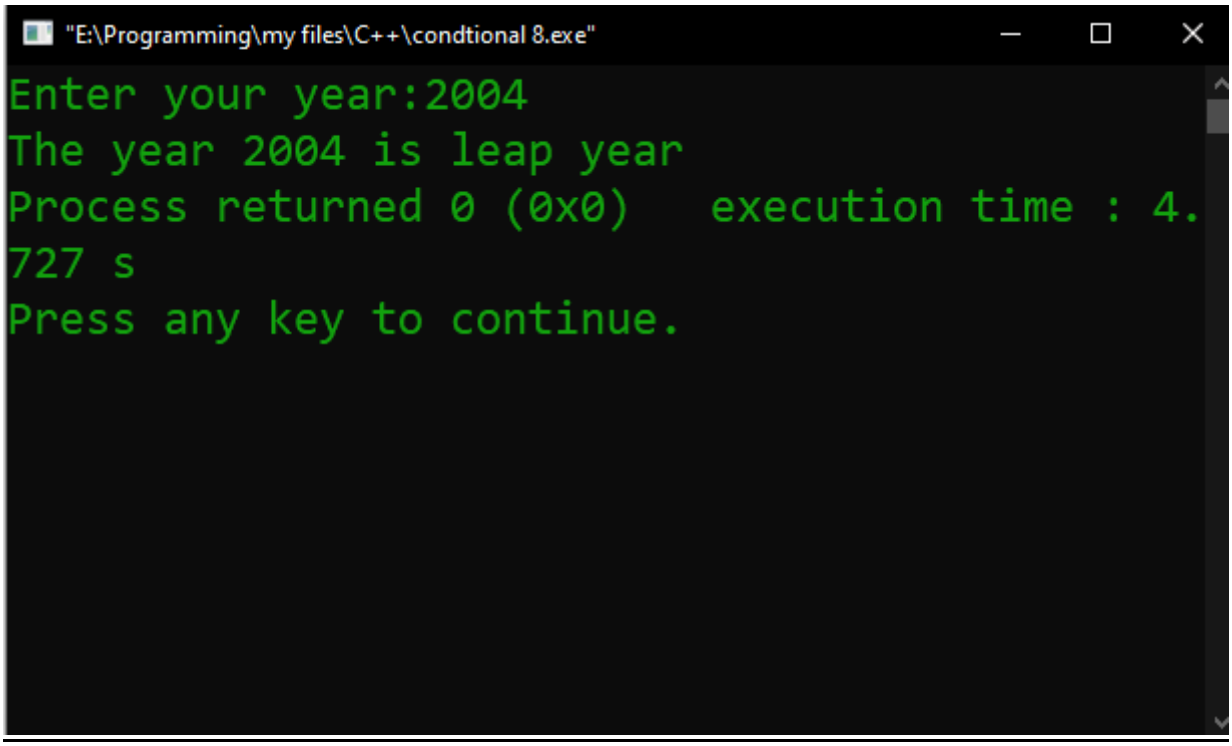
Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//check leap year using Conditional operator
#include <stdio.h>
int main()
{  int x;
   printf("Enter your year:");
   scanf("%d",&x);
     (x%4==0 && x%100!=0) || (x%400==0) ? printf("the year %d is leap year",x):printf("the year %d is a common year",x);
   return 0;
}
```

Output:



Output of leap year

Problem No. 09

Produce a code yourself to have the clear concept of comma operator, bitwise operator

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include <stdio.h>

int main()

{

int a,b;

printf("Enter your 2 numbers:\n");

  scanf("%d %d",&a,&b);

   //a=12,23,34;
    //b(12,23,34);

  printf("And operation of 2 numbers %d\n",a&b);
    printf("OR operation of 2 numbers %d\n",a||b);

   return 0;

}
```

Output:

```
"E:\Programming\my files\C++\condtional 9.exe"
Enter your 2 numbers:
15
45
And operation of 2 numbers 13
OR operation of 2 numbers 1

Process returned 0 (0x0)   execution time : 7.
813 s
Press any key to continue.
```

output

Lab Report 04

Problem No. 01

Name of the problem:

Take values of length and breadth of a rectangle from user and check if it is square or not.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h>
int main ()
{    int
x,y;
   printf("Enter your length of the rectangle:");
scanf("%d",&x);
   printf("Enter your length of the rectangle:");
scanf("%d",&y);
  if(x==y)
      printf("This is a Square");
else if(x>y || y>x)
      printf("This is a rectangle");
else
      printf("your input is in valid");
return 0;
}
```

Output:

```
"E:\Programming\my files\C++\decision 1.exe"
Enter your length of the rectangle:54
Enter your length of the rectangle:48
This is a rectangle
Process returned 0 (0x0)   execution time : 53
.852 s
Press any key to continue.
```

Figure 1 Output

Problem No. 02

Name of the problem:

Take two int values from user and print greatest among them.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int

main ()

{    int

x,y;

   printf("Enter Any two numbers:\n");    scanf("%d

%d",&x,&y);

   if(x>y)

      printf("The greatest number is %d",x);    else if(y>x)

      printf("The greatest number is %d",y);    else

      printf("Those number are equal");    return 0;

}
```

Output:

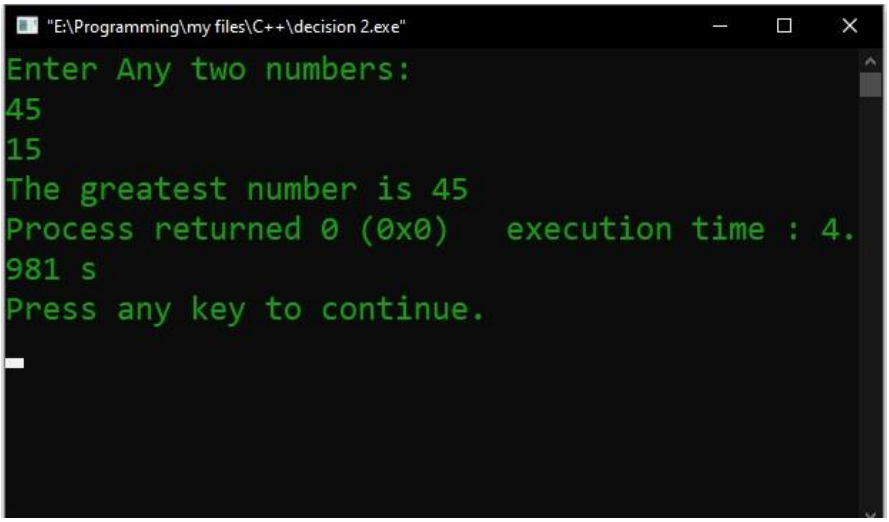


Figure 2 Output

Problem No. 03

Name of the problem:

Write a C program to input amount from user and print minimum number of notes (TK. 1000, 500, 100, 50, 20, 10, 5, 2, 1) required for the amount.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int
main ()

{

   int a,b=0,c=0,d=0,e=0,f=0,g=0,h=0,i=0,j=0,k=0;

   printf("notes of taka: TK. 1000,500,100,50,20,10,5,2,1\n");
printf("Enter the amount:");     scanf("%d",&a);

    printf("Total number of notes:");     if(a>=1000)

   {

      b=a/1000;

      a=a%1000;

   }

   if(a>=500)

   {

      c=a/500;

      a=a%500;

   }

   if(a>=200)

   {

      d=a/200;

      a=a%200;

   }

if(a>=100)

   {
```

```
  e=a/100;
      a=a%100;
   }
    if(a>=50)
   {
       f=a/50;
a=a%50;
   }
    if(a>=20)
   {
       g=a/20;
a=a%20;
   }
    if(a>=10)
   {
       h=a/10;
a=a%10;
   }
    if(a>=5)
   {
i=a/5;
       a=a%5;
   }
    if(a>=2)
   {
j=a/2;
       a=a%2;
   }
    if(a>=1)
   {
       k=a/1;
       a=a%1;
   }
   printf("\n 1000:%d \n 500:%d \n 200:%d \n 100:%d \n 50:%d \n 20:%d \n 10:%d \n 5:%d \n
2:%d \n 1:%d",b,c,d,e,f,g,h,i,j,k);
```

```
return 0; }
```

Output:

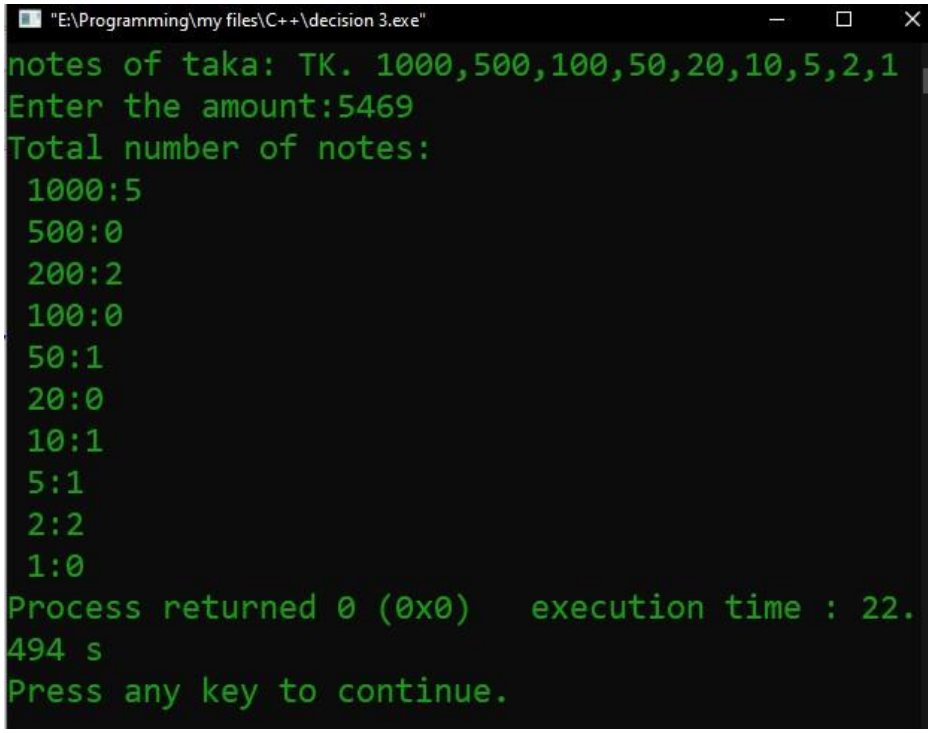


Figure 3 Output

Problem No. 04

Name of the problem:

A shop will give discount of 10% if the cost of purchased quantity is more than 1000. Ask user for quantity suppose, one unit will cost 100. Judge and print total cost for user.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int
main ()

{    int
x,y;

   printf("Enter the amount of your quantity:");
scanf("%d",&y);     x= y*100;     if(x>1000)

   {

      y= (x*10)/100;

      y= x-y;

      printf("Discount cost is %d",y);

   }

   else

      printf("You have no Discount..\n Total cost %d",x);     return 0;

}
```

Output:

```
"E:\Programming\my files\C++\decision 4.exe"
Enter the amount of your quantity:15
Discount cost is 1350
Process returned 0 (0x0)   execution time : 6.
879 s
Press any key to continue.
```

Figure 4 Output

Problem No. 05

Name of the problem

A company decided to give bonus of 5% to employee if his/her year of service is more than 5 years. Ask user for their salary and year of service and print the net bonus amount.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int
main ()
{
   int x,y,z;
   printf("Enter your salary:");     scanf("%d",&x);
   printf("Enter year of service:");
scanf("%d",&y);     if(y>5)
   {
     z=(x*5)/100;
z=x+z;
     printf("The bonus amount of employee is: %d",z);
   }
   Else
     printf("Sorry you have no bonus...");     return 0;}
```

Output:

```
"E:\Programming\my files\C++\decision 5.exe"
Enter your salary:45000
Enter year of service:6
The bonus amount of employee is: 47250
Process returned 0 (0x0)   execution time : 11
.360 s
Press any key to continue.
```

Figure 5 Output

Problem No. 06

Take input of age of 3 people by user and determine oldest and youngest among them.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program;

```
#include<stdio.h> int
main ()

{

   int x,y,z;

   printf("Enter the 1st age:\n");     scanf("%d",&x);

   printf("Enter the 2nd age:\n");     scanf("%d",&y);

   printf("Enter the 3rd age:\n");
scanf("%d",&z);     if(x>y && x>z)

      {printf("1st person is Oldest person\n");

      (y<z)? printf("2nd person is youngest\n"):printf("3rd person is youngest\n");}     else if(y>x && y>z)

      {printf("2nd person is Oldest person\n");

      (x<z)? printf("1st person is youngest\n"):printf("3rd person is youngest\n");}     else if(z>x && z>y)

      {printf("3rd person is Oldest person\n");

      (x<y)? printf("1st person is youngest\n"):printf("2nd person is youngest\n");}     else

      printf("Those number are equal");     return 0;

}
```

Output:

```
"E:\Programming\my files\C++\decision 6.exe"
Enter the 1st age:45
Enter the 2nd age:15
Enter the 3rd age:40
1st person is Oldest person
2nd person is youngest

Process returned 0 (0x0)   execution time : 10
.940 s
Press any key to continue.
```

Output 6

Problem no. 07

Write a C program to print absolute value of a number entered by user.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int
main ()
{     int x,y;     printf("INPUT:");
scanf("%d",&x);     if(x<0)
printf("OUTPUT:%d",x*-1);     else
       printf("OUTPUT:%d",x);
return 0;
}
```

Output:

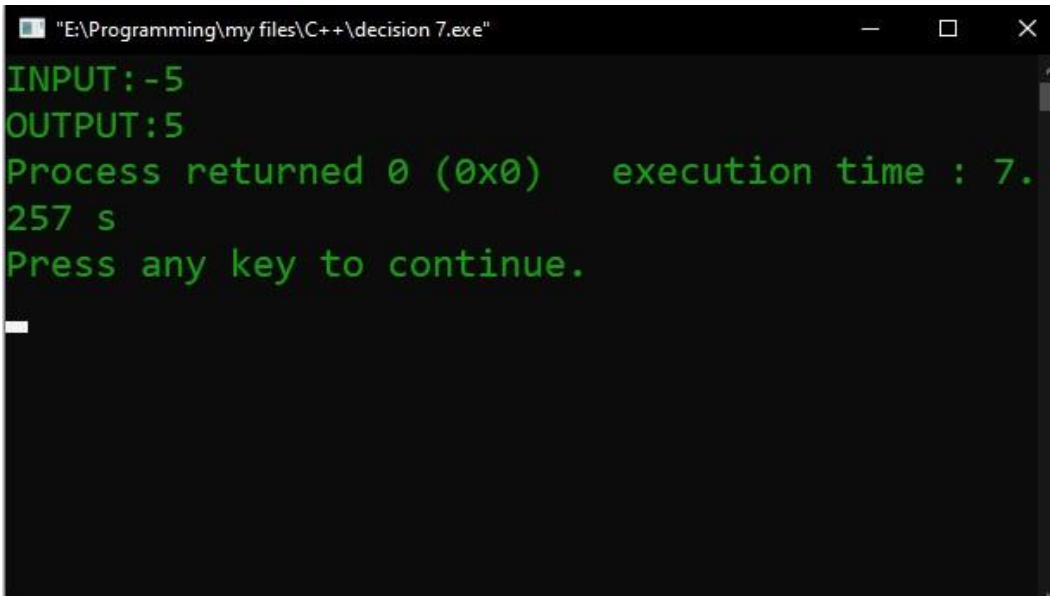


Figure 6

Problem No. 08

A student will not be allowed to sit in exam if his/her attendance is less than 75%.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
 #include<stdio.h>

int main ()

{

   int x,y,z;

   printf("Number of classes held:");     scanf("%d",&x);

   printf("Number of classes attented:");
scanf("%d",&y);     z=((y*100)/x);

   printf("Your attendance is %d%c\n",z,37);     if(z>=75)

      printf("You will be allowed to sit in exam\n\n");     else

      printf("You will not be allowed to sit in exam.\nBecause, your attendance is less then

75%c\n\n",37);

   return 0;

}
```

```
"E:\Programming\my files\C++\decision 8.exe"
Number of classes held:80
Number of classes attented:40
Your attendance is 50%
You will not be allowed to sit in exam.
Because, your attendance is less then 75%


Process returned 0 (0x0)   execution time : 5.
069 s
Press any key to continue.
```

```
"E:\Programming\my files\C++\decision 8.exe"
Number of classes held:80
Number of classes attented:69
Your attendance is 86%
You will be allowed to sit in exam


Process returned 0 (0x0)   execution time : 4.
894 s
Press any key to continue.
```

Problem No. 09

Write a C program to check if a year is leap year or not.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h>


int main ()

{
int y;

   printf("Enter year:");

   scanf("%d",&y);


   if(y%4==0)

      printf("This year %d is Leap year",y);     else
if(y==2000 || y==1900 || y==2100)

      (y%400==0)? printf("This year %d is Leap year",y) : printf("This year %d is not Leap year",y);     else

      printf("This year %d is not a Leap year",y);


   return 0;



}
```

```
"E:\Programming\my files\C++\decision 9.exe"
Enter year:2001
This year 2001 is not a Leap year
Process returned 0 (0x0)   execution time : 5.
228 s
Press any key to continue.
```

output

Problem No. 10

Admission to a professional course in subject to the following conditions: a) Marks in mathematics >=60

b) Marks in Physics >=50

c) Marks in Chemistry >=40

d) Total in all three subjects >=200 or
Total in mathematics and physics>=150.

Given the marks in the three subjects, write a program to process the applications to the eligible candidates.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
#include<stdio.h> int
main ()

{

   int m,p,c,total;     printf("Mark
in math:");     scanf("%d",&m);
printf("Mark in physics:");
scanf("%d",&p);

   printf("Mark in chemistry:");
scanf("%d",&c);     total= m+p+c;

   printf("Your total marks is:%d\n",total);     if(m>=60 &&
p>=50 && c>=40 && total>=200)        printf("You are
eligible candidates");     else

       printf("You are not eligible")     return 0; }}
```

Output:

```
"E:\Programming\my files\C++\decision 10.exe"
Mark in math:45
Mark in physics:48
Mark in chemistry:55
Your total marks is:148
You are not eligible
Process returned 0 (0x0)   execution time : 6.
663 s
Press any key to continue.
```

Lab Report 05

Problem No. 01

Name of the problem:

Implement the following problems in one code using switch-case, decision making & Conditional operator in C Language: using lab code 1-4. Lab will determinate as switch & problems will determinate as case value of the switch.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure. Here all the spaces are ignored in this code

Program:

```
//Lab report 05

//All Code till now

#include <stdio.h>

#include <conio.h>

int main()

{

   int l;

   printf("\tLab report Menu\n");

   printf("1. Lab-1 Problems\n2. Lab-2 Problems\n3. Lab-3 Problems\n4. Lab-4
Problems\n");

   printf("Enter your Lab Number:");

   scanf("%d",&l);
```

```
printf("\n");

switch(l)
{
   case 1:
   {
      int p;
printf("\tProblems Menu of Lab-1\n");
printf("Problem No.1\n");
printf("Problem No.2\n");
printf("Problem No.3\n");
printf("Enter your Problem No:");
scanf("%d",&p);
printf("\n");
      switch(p)
      {
         case 1:
         {
printf("Solving Problem 1\n");
printf("\tHello World\n");}
         break;
        case 2:
         {
printf("\tSolving Problem 2\n");
int x, y, sum;
```

```
    printf("Enter your first number:");
    scanf("%d", &x);
    printf("Enter your 2nd number:");
    scanf("%d", &y);
    sum=x+y;
    printf("The summation is:%d\n",sum);
}
            break;
           case 3:
            {
    printf("\tSolving Problem 3\n");
    int x, y, sub;
    printf("Enter your first number:");
    scanf("%d", &x);
    printf("Enter your 2nd number:");
    scanf("%d", &y);
    sub=x-y;
    printf("The summation is:%d\n",sub);
}
            break;
            default:
    printf("\tInvalid Problem Menu");
       break;
}
}
```

```
break;
   case 2:
{
     int p;
   printf("\tProblems Menu of Lab-2\n");
   printf("Problem No.1\n");
   printf("Problem No.2\n");
   printf("Problem No.3\n");
   printf("Problem No.4\n");
   printf("Problem No.5\n");
   printf("Problem No.6\n");
   printf("Enter your Problem No:");
   scanf("%d",&p);
   printf("\n");
   switch(p)
{
   case 1:
{
    printf("\tSolving Problem 1\n");
   float x, y, div;
   printf("Enter your 2 numbers:\n");
   scanf("%f %f", &x, &y);
   div=x/y;
   printf("Your division is: %f\n", div);
}
```

```
    break;
    case 3:
{
    printf("\tSolving Problem 3\n");
    int x;
    float c,f;
    printf("Converter!! \n press 1 for convert Celsius to Fahrenheit \n press 2 for
convert  Fahrenheit to Celsius \n");
    scanf("%d", &x);
    switch(x)
{
        case 1 :
        //float c, f;
        printf("Enter the Celsius value:");
        scanf("%f",&c);
        f=(1.8*c)+32;
        printf("The Fahrenheit value is:%f",f);
        break;

        case 2:
        //float f,c;
        printf("Enter the Fahrenheit value:");
        scanf("%f",&f);
        c=((f-32)*5)/9;
        printf("The Celsius value is:%f",c);
```

```
        break;
        default :
        printf("your value is not valid");
        break;
}
        break;
}
    case 2:
    {
        printf("\tSolving Problem 2\n");
         int x,h,e,a,b,t,r,s,c;
    printf("Enter 1 for area of triangle \n Enter 2 for are of rectangle \n Enter 3 for are of square \n Enter 4 for area of circle\n Enter 5 for area of cube\n");
    scanf("%d", &x);
    switch(x)
    {
        case 1 :
        printf("Enter the height value of the triangle : ");
        scanf("%d", &h);
        printf("Enter the edge value of the triangle: ");
        scanf("%d", &e);
        t=0.5*(h*e);
        printf("The area of the triangle is:%d",t);
        break;
        case 2:
```

```
printf("Enter the length value of the rectangle: ");
scanf("%d", &a);
printf("Enter the width value of the rectangle: ");
scanf("%d", &b);
r=a*b;
printf("The area of the rectangle is:%d",r);
break;
case 3:
printf("Enter the edge of the square : ");
scanf("%d", &a);
s=a*a;
printf("The area of the square is:%d", s);
break;

case 4:
printf("Enter the radius :\n");
scanf("%d", &a);
c=3.1416*(a*a);
printf("The area of the Circle is:%d", s);
break;
case 5:
printf("Enter the edge of the cube : ");
scanf("%d", &a);
s=a*a*a;
printf("The area of the cube is:%d", c);
```

```
        break;
        default :
        printf("your value is not valid");
}
}
    break;
    case 4:
    {
            printf("\tSolving Problem 4\n");
            int a, b;
            printf("Enter the value of A=");
            scanf("%d",&a);
            printf("enter the value of B=");
            scanf("%d",&b);
            a=a;
            b=b-a;
            a=a+b;

            printf("A=%d \nB=%d", a,b);
}
    break;
    case 5:
    {
            printf("\tSolving Problem 5\n");
             int x, y, z, p;
```

```
    printf("Enter number of people in the village last year:");
    scanf("%d", &x);
    printf("Enter the number of people moving in:");
    scanf("%d", &y);
    printf("Enter the number of people moving out:");
    scanf("%d", &z);
    p = (x + y) - z;
    printf("Number of people this year is : %d", p);
}
   break;
   case 6:
   {
       printf("\tSolving Problem No. 6\n");
        float paid, beef, mutton, change;
 printf("Amount being paid: ");
 scanf("%f", &paid);
 printf("Kilos of beef getting bought: ");
 scanf("%f", &beef);
 printf("Kilos of mutton getting bought: ");
 scanf("%f", &mutton);
 change = paid - (beef * 700 + mutton * 800);
 printf("Amount of change is : %f", change);
}
   break;
   default :
```

```
    printf("\tInvalid Problem Menu\n");
    break;
}
break;
    case 3:
    {
        int p;
    printf("\tProblems Menu of Lab-3\n");
    printf("Problem No.1\n");
    printf("Problem No.2\n");
    printf("Problem No.3\n");
    printf("Problem No.4\n");
    printf("Enter your Problem No:");
    scanf("%d",&p);
    printf("\n");
    switch(p)
    {
        case 1:
        {
            printf("\tSolving Problem 1\n");
            int x,y;
 printf("Enter your first number:");
 scanf("%d",&x);
 printf("Enter your second number:");
 scanf("%d",&y);
```

```
(x>y) ? printf("The largest number is:%d",x) : printf("The largest number is:%d",x);
        }
    break;
      case 2:
      {
          printf("\tSolving Problem 2\n");
          int x;
printf("Enter your first number:");
scanf("%d",&x);
(x>0) ? printf("The number is positive +%d",x) : printf("The number is negative %d",x);
      }
      break;
      case 3:
      {
          printf("\tSolving Problem 3\n");
          int x;
printf("Enter your first number:");
scanf("%d",&x);
(x%2==0) ? printf("The number is even: %d",x) : printf("The number is odd: %d",x);
      }
      break;
```

```
        case 4:
        {
            printf("\tSolving Problem 4\n");
            int item;
 printf("Enter the number of item:");
 scanf("%d",&item);
 printf("you have %d item%s",item,(item==1) ? "" : "s");
        }
        break;
    default :
    printf("\tInvalid Problem Menu\n");
    break;
}
        break;
    case 4:
{
        int p;
    printf("\tProblems Menu of Lab-4\n");
    printf("Problem No.1\n");
    printf("Problem No.2\n");
    printf("Problem No.3\n");
    printf("Problem No.4\n");
    printf("Problem No.5\n");
    printf("Problem No.6\n");
    printf("Problem No.7\n");
```

```
   printf("Problem No.8\n");
   printf("Problem No.9\n");
   printf("Problem No.10\n");
   printf("Enter your Problem No:");
   scanf("%d",&p);
   printf("\n");
  switch(p)
{
       case 1:
       {
       printf("\tSolving Problem 1\n");
       int x,y;
 printf("Enter your length of the rectangle:");
 scanf("%d",&x);
 printf("Enter your length of the rectangle:");
 scanf("%d",&y);
 if(x==y)
 printf("This is a Square");
 else if(x>y || y>x)
 printf("This is a rectangle");
 else
 printf("your input is in valid");
   }
   break;
   case 2:
```

```
      {
         printf("\tSolving Problem 2\n");
         int x,y;
printf("Enter Any two numbers:\n");
scanf("%d %d",&x,&y);
if(x>y)
printf("The greatest number is %d",x);
else if(y>x)
printf("The greatest number is %d",y);
else
printf("Those number are equal");
      }
   break;
     case 3:
      {
         printf("\tSolving Problem 3\n");
         int a,b=0,c=0,d=0,e=0,f=0,g=0,h=0,i=0,j=0,k=0;
printf("notes of taka: TK. 1000,500,100,50,20,10,5,2,1\n");
printf("Enter the amount:");
scanf("%d",&a);
printf("Total number of notes:");
if(a>=1000)
{
b=a/1000;
a=a%1000;
```

```
}
if(a>=500)
{
c=a/500;
a=a%500;
}
if(a>=200)
{
d=a/200;
a=a%200;
}
if(a>=100)
{
e=a/100;
a=a%100;
}
if(a>=50)
{
f=a/50;
a=a%50;
}
if(a>=20)
{
g=a/20;
a=a%20;
```

```
}
if(a>=10)
{
h=a/10;
a=a%10;
}
if(a>=5)
{
i=a/5;
a=a%5;
}
if(a>=2)
{
j=a/2;
a=a%2;
}
if(a>=1)
{
k=a/1;
a=a%1;
}
printf("\n 1000:%d \n 500:%d \n 200:%d \n 100:%d \n 50:%d \n 20:%d \n 10:%d
\n 5:%d \n 2:%d \n 1:%d",b,c,d,e,f,g,h,i,j,k);
  }
  break;
```

```
    case 4:
     {
        printf("\tSolving Problem 4\n");
        int x,y;
printf("Enter the amount of your quantity:");
scanf("%d",&y);
x= y*100;
if(x>1000)
{
y= (x*10)/100;
y= x-y;
printf("Discount cost is %d",y);
}
else
printf("You have no Discount..\n Total cost %d",x);
     }
  break;


     case 5:
     {
        printf("\tSolving Problem 5\n");
        int x,y,z;
printf("Enter your salary:");
scanf("%d",&x);
printf("Enter year of service:");
```

```
scanf("%d",&y);
if(y>5)
{
z=(x*5)/100;
z=x+z;
printf("The bonus amount of employee is: %d",z);
}
else
printf("Sorry you have no bonus...");
      }
   break;
      case 6:
      {
         printf("\tSolving Problem 6\n");
         int x,y,z;
printf("Enter the 1st age:\n");
scanf("%d",&x);
printf("Enter the 2nd age:\n");
scanf("%d",&y);
printf("Enter the 3rd age:\n");
scanf("%d",&z);
if(x>y && x>z)
{printf("1st person is Oldest person\n");
(y<z)? printf("2nd person is youngest\n"):printf("3rd person is youngest\n");}
else if(y>x && y>z)
```

```
{printf("2nd person is Oldest person\n");
(x<z)? printf("1st person is youngest\n"):printf("3rd person is youngest\n");}
else if(z>x && z>y)
{printf("3rd person is Oldest person\n");
(x<y)? printf("1st person is youngest\n"):printf("2nd person is youngest\n");}
else
printf("Those number are equal");
    }
  break;
      case 7:
      {
         printf("\tSolving Problem 7\n");
         int x,y;
printf("INPUT:");
scanf("%d",&x);
if(x<0)
printf("OUTPUT:%d",x*-1);
else
printf("OUTPUT:%d",x);
      }
   break;
      case 8:
      {
         printf("\t\nSolving Problem 8");
         int x,y,z;
```

```
printf("Number of classes held:");
scanf("%d",&x);
printf("Number of classes attented:");
scanf("%d",&y);
z=((y*100)/x);
printf("Your attendance is %d%c\n",z,37);
if(z>=75)
printf("You will be allowed to sit in exam\n\n");
else
printf("You will not be allowed to sit in exam.\nBecause, your attendance is less then 75%c\n\n",37);
        }
    break;


        case 9:
        {
            printf("\tSolving Problem 9\n");
            int y;
printf("Enter year:");
scanf("%d",&y);
if(y%4==0)
printf("This year %d is Leap year",y);
else if(y==2000 || y==1900 || y==2100)
(y%400==0)? printf("This year %d is Leap year",y) : printf("This year %d is not Leap year",y);
```

```
else
printf("This year %d is not a Leap year",y);
  }
  break;
     case 10:
     {
        printf("\tSolving Problem 10\n");
        int m,p,c,total;
printf("Mark in math:");
scanf("%d",&m);
printf("Mark in physics:");
scanf("%d",&p);
printf("Mark in chemistry:");
scanf("%d",&c);
total= m+p+c;
printf("Your total marks is:%d\n",total);
if(m>=60 && p>=50 && c>=40 && total>=200)
printf("You are eligible candidates");
else
printf("You are not eligible");
     }
     break;
  default :
  printf("\tInvalid Problem Menu\n");
  return 0;
```

```
}
}
}
}
    break;
    default :
    printf("\tInvalid lab Menu\n");
    break;
}
        return 0;
}
```

OutPut:

```
        Lab report Menu
1. Lab-1 Problems
2. Lab-2 Problems
3. Lab-3 Problems
4. Lab-4 Problems
Enter your Lab Number:2

        Problems Menu of Lab-2
Problem No.1
Problem No.2
Problem No.3
Problem No.4
Problem No.5
Problem No.6
Enter your Problem No:6

        Solving Problem No. 6
Amount being paid: 5000
Kilos of beef getting bought: 2
Kilos of mutton getting bought: 1
Amount of change is : 2800.000000
Process returned 0 (0x0)   execution time : 10.809 s
Press any key to continue.
```

output 15

```
        Lab report Menu
1. Lab-1 Problems
2. Lab-2 Problems
3. Lab-3 Problems
4. Lab-4 Problems
Enter your Lab Number:4

        Problems Menu of Lab-4
Problem No.1
Problem No.2
Problem No.3
Problem No.4
Problem No.5
Problem No.6
Problem No.7
Problem No.8
Problem No.9
Problem No.10
Enter your Problem No:10

        Solving Problem 10
Mark in math:60
Mark in physics:78
Mark in chemistry:80
Your total marks is:218
You are eligible candidates
Process returned 0 (0x0)   execution time : 12.259 s
Press any key to continue.
```

output 2

```
        Lab report Menu
1. Lab-1 Problems
2. Lab-2 Problems
3. Lab-3 Problems
4. Lab-4 Problems
Enter your Lab Number:4

        Problems Menu of Lab-4
Problem No.1
Problem No.2
Problem No.3
Problem No.4
Problem No.5
Problem No.6
Problem No.7
Problem No.8
Problem No.9
Problem No.10
Enter your Problem No:8


Solving Problem 8
Number of classes held:32
Number of classes attented:20
Your attendance is 62%
You will not be allowed to sit in exam.
Because, your attendance is less then 75%


Process returned 0 (0x0)   execution time : 7.957 s
Press any key to continue.
```

output 4

```
        Lab report Menu
1. Lab-1 Problems
2. Lab-2 Problems
3. Lab-3 Problems
4. Lab-4 Problems
Enter your Lab Number:4

        Problems Menu of Lab-4
Problem No.1
Problem No.2
Problem No.3
Problem No.4
Problem No.5
Problem No.6
Problem No.7
Problem No.8
Problem No.9
Problem No.10
Enter your Problem No:45

        Invalid Problem Menu

Process returned 0 (0x0)   execution time : 6.000 s
Press any key to continue.
```

output 5

Lab Report 06

Problem No. 01

Name of the problem:

Write a C program to find the Winner of Rock Paper Scissor Game Among Two Users

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Rock Paper Scissor Game
#include <stdio.h>
int main()
{
   char p1,p2;
   printf("\tLets play rock paper scissor.\n\n");
   printf("Player 1 select your Choice.\nAmong Rock, Paper, Scissor\n");
   printf("Press 'r' for Rock\nPress 'p' for Paper\nPress 's' for Scissor\n");
   scanf("%c",&p1);
   printf("Player 2 select your Choice");
   printf("\nPress 'r' for Rock\nPress 'p' for Paper\nPress 's' for Scissor\n");
   scanf(" %c",&p2);
   switch(p1)
   {
   case 'r':
```

```
    (p2=='r')? printf("Game Draw!!") : (p2=='s')? printf("Player 1 Win...!!") : printf("Player
2 Win...!!");
  break;
  case 's':
    (p2=='s')? printf("Game Draw!!") : (p2=='p')? printf("Player 1 Win...!!") : printf("Player
2 Win...!!");
  break;
  case 'p':
    (p2=='p')? printf("Game Draw!!") : (p2=='r')? printf("Player ! Win...!!") : printf("Player
2 Win...!!");
  break;
  default :
    printf("Game Error");
  break;
  }
}
```

Output :

```
        Lets play rock paper scissor.

Player 1 select your Choice.
Among Rock, Paper, Scissor
Press 'r' for Rock
Press 'p' for Paper
Press 's' for Scissor
r
Player 2 select your Choice
Press 'r' for Rock
Press 'p' for Paper
Press 's' for Scissor
p
Player 2 Win...!!


Process returned 0 (0x0)   execution time : 3.902 s
Press any key to continue.
```

Lab Report 07

Problem No. 01

Name of the problem:

Write a C program using a switch case to check whether the number entered by the user is odd or even.

Introduction

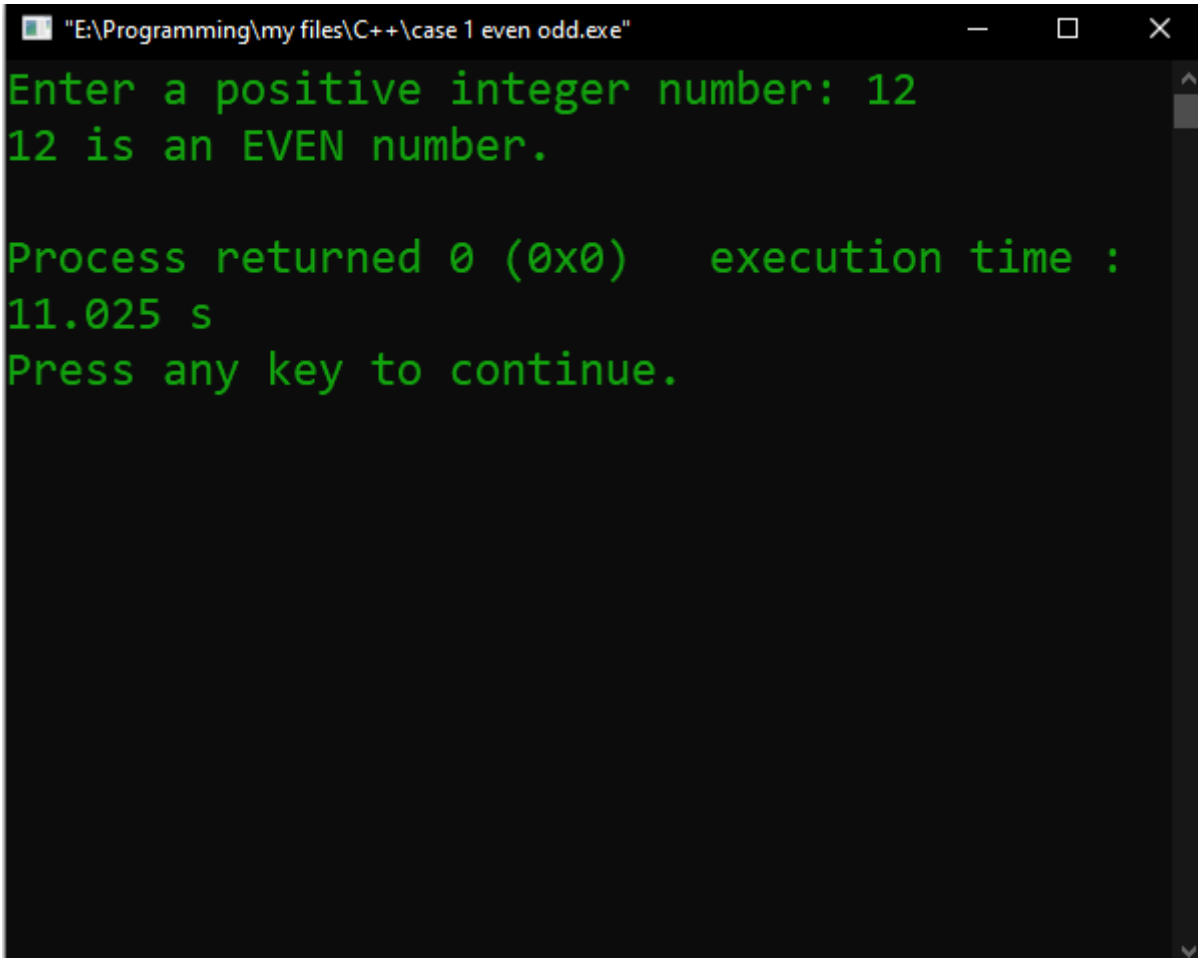
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//check whether the number entered by the user is odd or even using switch case
#include <stdio.h>
int main()
{
 int number;
printf("Enter a positive integer number: ");
scanf("%d",&number);
 switch(number%2) //this will return either 0 or 1
 {
 case 0:
 printf("%d is an EVEN number.\n",number);
 break;
 case 1:
 printf("%d is an ODD number.\n",number);
```

```
 break;
 }
 return 0;
}
```

Output :



Problem No. 02

Name of the problem:

Write a C program to checks whether a character entered by the user is a vowel or not by using the switch case statement.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//checks whether a character entered by the user is a vowel or not by using the switch case statement

#include <stdio.h>

int main()

{

   char ch;

   printf("Enter any alphabet: ");

   scanf("%c", &ch);

 switch(ch)

 {

 case 'a':

 printf("Entered Alphabet is Vowel");

 break;

 case 'e':

 printf("Entered Alphabet is Vowel");

 break;
```

```
case 'i':
printf("Entered Alphabet is Vowel");
break;
case 'o':
printf("Entered Alphabet is Vowel");
break;
case 'u':
printf("Entered Alphabet is Vowel");
break;
case 'A':
printf("Entered Alphabet is Vowel");
break;
case 'E':
printf("Entered Alphabet is Vowel");
break;
case 'I':
printf("Entered Alphabet is Vowel");
break;
case 'O':
printf("Entered Alphabet is Vowel");
break;
case 'U':
printf("Entered Alphabet is Vowel");
break;
default:
printf("Consonant");
}
```

```
 return 0;
}
```

Output :

Problem No. 03

Name of the problem:

Write a menu-driven C program to find the area of the circle, area of the triangle and area of the rectangle according to the user's input choice

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//finding area of circle, rectangle, triangle
#include <stdio.h>
int main()
{
   printf("\tMenu of shapes\n");
   printf("Select your shapes:\n");
   printf("1.Area of circle.\n2.Area of reactangle.\n3.Area of tringle.\n\n");
   printf("Enter your choice:");
   int m,breadth,length,h,e;
   float radius,area;
   scanf("%d",&m);
   switch(m)
{
case 1:
   {
   printf("\tArea of circle\n");
   printf("Enter the radius:");
```

```
   scanf("%f",&radius);
   area = 3.142*radius*radius;
   printf("Area of a circle = %f\n", area);
   }
break;
case 2:
   {
   printf("\tArea of rectangle\n");
   printf("Enter the breadth and length:");
   scanf("%d %d", &breadth, &length);
   area = breadth*length;
   printf("Area of a Reactangle = %f\n", area);
   }
break;
case 3:
   {
   printf("\tArea of triangle\n");
   printf("Enter the height value of the triangle :");
   scanf("%d", &h);

   printf("Enter the edge value of the triangle :");
   scanf("%d", &e);

   area=0.5*(h*e);
   printf("The area of the triangle is:%d", area);
   }
Break; default: printf("\tInvalid Menu"); break;}}
```

Problem No. 04

Name of the problem:

Write a menu-driven C program to convert a given year into, Months Days Hours Minutes Seconds. For simplicity don't include leap year, use 1 Year = 365 days 1 Month = 30 days

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//convert a given year into month,day
#include<stdio.h>
int main()
{
 int m,years,months,days,hours,minutes,seconds;
 printf("Select your choice to convert a year into:\n");
 printf("1.Month\n2.Days\n3.Hours\n4.Minutes\n5.Seconds\n");
 printf("Enter your choice:");
 scanf("%d",&m);
 printf("\nEnter the number of years:");
 scanf("%d",&years);
 switch(m)
 {
 case 1:
   {
      months = 12 * years;
      printf("\n%d years = %d Months",years,months);
```

```
  }
break;
case 2:
  {
     days = 365 * years;
     printf("\n%d years = %d Days",years,days);
  }
break;
case 3:
  {
     hours = 24 * 365 * years;
     printf("\n%d years = %d Hours",years,hours);
  }
break;
case 4:
  {
      minutes = 60 * 24 * 365 * years;
      printf("\n%d years = %d minutes",years,minutes);
  }
break;
case 5:
  {
     seconds = 60 * 60 * 24 * 365 * years;
     printf("\n%d years = %d seconds",years,seconds);
  }
default:
  printf("Invalid Menu");} return 0;}
```

Bonus Problem

Name of the problem:

A computer showroom has announced the following seasonal discounts on purchase of items: Purchase Amount Discount Pen Drive RAM 0-100 --- 5% 101-200 5% 7.5% 201-300 7.5% 10.0% Above300 10.0% 15.0% Write a program using switch and if statements to compute the net amount to be paid by a customer

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Shop For Pendrive & Ram
 #include<stdio.h>
 int main()
 {
 int amount,price,ch;
 float net,discount,amountToBePaid;
 printf("\tShop\n1.Pendrive purchase\n2.Ram purchase\n");
 printf("Enter your choice");
 scanf("%d",&ch);
 printf("\n Enter purchased amount: ");
 scanf("%d",&amount);
 printf("Enter the price per item: ");
 scanf("%d", &price);
 net = amount * price;
 switch(ch)
{
```

```
case 1:
if(amount<=100)
{
amountToBePaid = net;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>=101 && amount<=200)
{
discount = (net*5)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>=201 && amount<=300)
{
discount = (net*7.5)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>300)
{
discount = (net*10)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
break;
case 2:
if(amount<=100)
```

```
{
discount = (net*5)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>=101 && amount<=200)
{
discount = (net*7.5)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>=201 && amount<=300)
{
discount = (net*10)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
else if(amount>300)
{
discount = (net*15)/100;
amountToBePaid = net-discount;
printf("After applying the discount, you have to pay: %0.2f", amountToBePaid);
}
break;
}
getch();
}
```

Lab Report 08

Problem No. 01

Name of the problem:

Write a program in C to display n terms of natural number and their sum.

Test Data : 7

Expected Output :

The first 7 natural number is : 1 2 3 4 5 6 7

The Sum of Natural Number upto 7 terms : 28

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Displaying the numbers 1 to 10 using for loop
#include <stdio.h>
int main()
{
  int i,n,sum=0;
  printf("Input Value of terms : ");
  scanf("%d",&n);
  printf("\nThe first %d natural numbers are:\n",n);
  for(i=1;i<=n;i++)
  {
   printf("%d ",i);
   sum+=i;
  }
```

```
  printf("\nThe Sum of natural numbers upto %d terms : %d \n",n,sum);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 1.exe"
Input Value of terms : 5

The first 5 natural numbers are:
1 2 3 4 5
The Sum of natural numbers upto 5 terms : 15

Process returned 0 (0x0)   execution time : 4.5
32 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to read 10 numbers from keyboard and find their sum and average.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Display the average of the values loop
#include <stdio.h>
int main()
{
   int i,n,sum=0;
        float avg;
        printf("Input the 10 numbers : \n");
        for (i=1;i<=10;i++)
        {
           printf("Number-%d :",i);
                scanf("%d",&n);
                sum +=n;
        }
        avg=sum/10.0;
        printf("The sum of 10 no is : %d\nThe Average is : %f\n",sum,avg);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 2.exe"
Input the 10 numbers :
Number-1 :1
Number-2 :2
Number-3 :3
Number-4 :4
Number-5 :5
Number-6 :6
Number-7 :7
Number-8 :8
Number-9 :9
Number-10 :10
The sum of 10 no is : 55
The Average is : 5.500000

Process returned 0 (0x0)   execution time : 10
```

Problem No. 03

Name of the problem:

Write a program in C to display the cube of the number upto given an integer.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the cube of the number up to given an integer
#include <stdio.h>
int main()
{
  int i,n,c=0;
            float avg;
            printf("Input number of terms : \n");
            scanf("%d",&n);
            for (i=1;i<=n;i++)
            {
             c=i*i*i;
             printf("Number is : %d and cube of the %d is :%d\n",i,i,c);
            }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 3.exe"
Input number of terms :
5
Number is : 1 and cube of the 1 is :1
Number is : 2 and cube of the 2 is :8
Number is : 3 and cube of the 3 is :27
Number is : 4 and cube of the 4 is :64
Number is : 5 and cube of the 5 is :125

Process returned 0 (0x0)   execution time : 3.
464 s
Press any key to continue.
```

Problem No. 04

Name of the problem:

Write a program in C to display the multiplication table of a given integer.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Table of inputed number
#include<stdio.h>
int main()
{
   int n,i,t=0;
   printf("Enter your n:");
   scanf("%d",&n);
   for(i=1;i<=10;i++)
   {
      t=n*i;
      printf("%d * %d = %d\n",n,i,t);
   }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\lab_loop_4.exe"
Enter your n:3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

Problem No. 05

Name of the problem:

Write a program in C to display the multiplication table vertically from 1 to n.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Table of nth inputed number
#include<stdio.h>
int main()
{
   int n,i,t=0,x;
   printf("Enter your n:");
   scanf("%d",&n);

   for(i=1;i<=n;i++)
   {
      printf("\ntable of %d\n",i);
      for(x=1;x<=10;x++)
      {
         t=x*i;
         printf("%d * %d = %d\n",x,i,t);
      }
   }
```

```
    return 0;
}
```

Output

```
"E:\Programming\my files\C++\looping\lab_loop_4.exe"
Enter your n:3
3 * 1 = 3
3 * 2 = 6
3 * 3 = 9
3 * 4 = 12
3 * 5 = 15
3 * 6 = 18
3 * 7 = 21
3 * 8 = 24
3 * 9 = 27
3 * 10 = 30
```

Problem No. 06

Name of the problem:

Write a program in C to display the n terms of odd natural number and their sum

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the n terms of odd natural number and their sum
#include <stdio.h>
int main()
{
  int i,n,sum=0;


  printf("Input number of terms : ");
  scanf("%d",&n);
  printf("\nThe odd numbers are :");
  for(i=1;i<=n;i++)
  {
    printf("%d ",2*i-1);
    sum+=2*i-1;
  }
  printf("\nThe Sum of odd Natural Number upto %d terms : %d \n",n,sum);
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 6.exe"
Input number of terms : 10

The odd numbers are :1 3 5 7 9 11 13 15 17 19
The Sum of odd Natural Number upto 10 terms : 100

Process returned 0 (0x0)   execution time : 1.992 s
Press any key to continue.
```

Problem No. 07

Name of the problem:

Write a program in C to display the pattern like right angle triangle using an asterisk.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//* star looping triangle *-*
#include <stdio.h>
int main()
{
   int i,j;
   int n;
   printf("Enter triangle line numbers:");
   scanf("%d",&n);

   for(i=1; i<=n; i++)
   {
      for(j=1; j<=i; j++)
      printf("*");
      printf("\n");
   }
   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\looping\lab_loop_7.exe"
Enter triangle line numbers:7
*
**
***
****
*****
******
*******

Process returned 0 (0x0)   execution time : 3.
660 s
Press any key to continue.
```

Problem No. 08

Name of the problem:

Write a program in C to display the pattern like right angle triangle with a number

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//display the pattern like right angle triangle with a number
#include <stdio.h>
int main()
{
  int i,j,rows;
  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",j);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 8.exe"
Input number of rows : 8
1
12
123
1234
12345
123456
1234567
12345678

Process returned 0 (0x0)   execution time : 1.
995 s
```

Problem No. 09

Name of the problem:

Write a program in C to make such a pattern like right angle triangle with a number which will repeat a number in a row

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like right angle triangle with a number which will repeat a number in a row.
#include <stdio.h>
void main()
{
  int i,j,rows;


  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",i);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 9.exe"
Input number of rows : 9
1
22
333
4444
55555
666666
7777777
88888888
999999999

Process returned 9 (0x9)   execution time : 0.
```

Problem No. 10

Name of the problem:

Write a program in C to make such a pattern like right angle triangle with number increased by 1.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like right angle triangle with a number which will repeat a number in a row.
#include <stdio.h>
void main()
{
  int i,j,rows;


  printf("Input number of rows : ");
  scanf("%d",&rows);
  for(i=1;i<=rows;i++)
  {
            for(j=1;j<=i;j++)
              printf("%d",i);
            printf("\n");
  }
}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 10.exe"
Input number of rows : 6
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
16 17 18 19 20 21

Process returned 0 (0x0)   execution time : 5.
773 s
Press any key to continue.
```

Problem No. 11

Name of the problem:

Write a program in C to make such a pattern like a pyramid with numbers increased by 1.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//pattern like a pyramid with numbers increased by 1.
#include <stdio.h>
int main()
{
  int i,j,spc,rows,k,t=1;
  printf("Input number of rows : ");
  scanf("%d",&rows);
  spc=rows+4-1;
  for(i=1;i<=rows;i++)
  {
      for(k=spc;k>=1;k--)
        {
          printf(" ");
        }
                for(j=1;j<=i;j++)
                printf("%d ",t++);
              printf("\n");
```

```
spc--;
}}
```

Output:

```
"E:\Programming\my files\C++\looping\loop 11.exe"
Input number of rows : 4
       1
      2 3
     4 5 6
    7 8 9 10

Process returned 0 (0x0)   execution time : 0.
708 s
Press any key to continue.
```

Lab Report 09

Problem No. 01

Name of the problem:

Write a program in C to store elements in an array and print it.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//print array number
#include<stdio.h>
int main()
{
   int a[4],i;
   printf("Enter 5 numbers\n");

   for(i=1;i<=5;i++)
   scanf("%d",&a[i]);

   for(i=1;i<=5;i++)
   printf("a[%d]=%d\n",i,a[i]);

   return 0;}
```

Output:

```
"E:\Programming\my files\C++\array\array_1.exe"
Enter 5 numbers
1 2 3 4 5
a[1]=1
a[2]=2
a[3]=3
a[4]=4
a[5]=5

Process returned 0 (0x0)   execution time : 3.
786 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to read n number of values in an array and display it in reverse order

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//print reverse array numbers
#include<stdio.h>
int main()
{
   int a[5],i;
   printf("\tEnter 5 numbers\n");

   for(i=1;i<=5;i++)
            scanf("\t%d",&a[i]);

      printf("\tOutput\n");

   for(i=5;i>=1;i--)
   printf("\ta[%d]=%d\n",i,a[i]);

   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\array\array_2.exe"
        Enter 5 numbers
1
2
3
4
5
        Output
        a[5]=5
        a[4]=4
        a[3]=3
        a[2]=2
        a[1]=1

Process returned 0 (0x0)   execution time : 4.
609 s
Press any key to continue.
```

Problem No. 03

Name of the problem:

Write a program in C to find the sum of all elements of the array.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Summation of array numbers
#include<stdio.h>
int main()
{
   int a[5],i,sum=0;
   printf("Enter 5 numbers\n");

   for(i=0;i<=4;i++)
   scanf("%d",&a[i]);

   for(i=0;i<5;i++)
      sum=sum+a[i];

   printf("The summation is: %d",sum);

   return 0;
}
```

Output:

```
"E:\Programming\my files\C++\array\array_3.exe"
Enter 5 numbers
1
2
3
4
5
The summation is: 15
Process returned 0 (0x0)   execution time : 3.
744 s
Press any key to continue.
```

Problem No. 04

Name of the problem:

Write a program in C to find the maximum and minimum element in an array.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//find Max & min value from array numbers
#include <stdio.h>
int main()
{
 int i,m,s;
 int a[5];
 printf("Enter 5 numbers of elements:\n");
 for (i=0; i<=4; ++i)
  {
  scanf("%d",&a[i]);
  }
  m=a[0];
  s=a[0];
 for (i=1; i<=4; ++i)
  {
  if (a[0]<a[i])
   {m=a[i];}
```

```
  if (a[0]>a[i])
    {s=a[i];}
 }
 printf("Maximum element = %d\n",m);
 printf("Minimum element = %d",s);
 return 0;
}
```

Output:

```
"E:\Programming\my files\C++\array\array_4.exe"
Enter 5 numbers of elements:
1
2
3
4
5
Maximum element = 5
Minimum element = 1
Process returned 0 (0x0)   execution time : 54
.826 s
Press any key to continue.
```

Problem No. 05

Name of the problem:

Write a program in C to sort elements of array in ascending order

Introduction

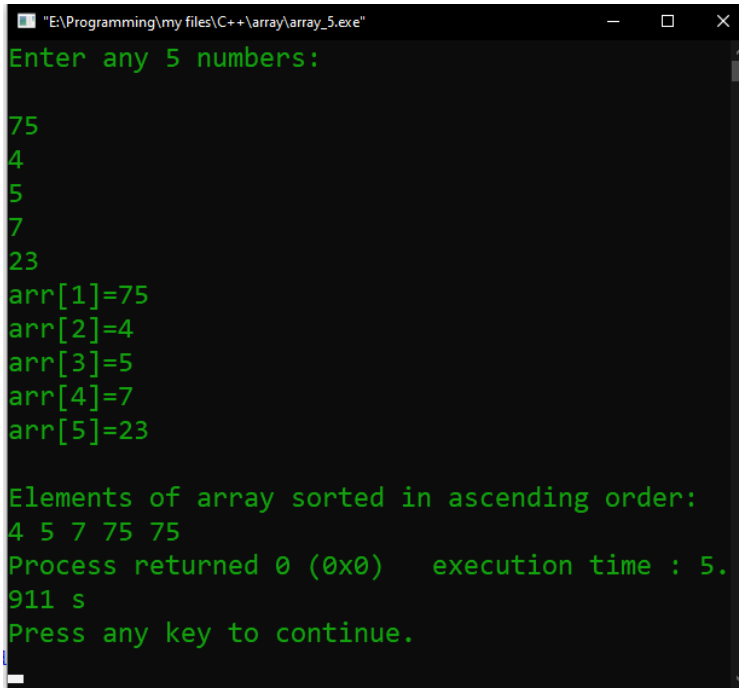
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//sort elements of array in ascending order
#include <stdio.h>
int main()
{
   int i,arr[5],j;
   int temp=0;
   printf("Enter any 5 numbers:\n");
   for(i=1;i<=5;i++)
   scanf("%d",&arr[i]);
   for(i=1;i<=5;i++)
   printf("arr[%d]=%d\n",i,arr[i]);
 //Sort the array in ascending order
   for (i=1; i<=5; i++)
   {
      for (j=i+1; j<=5; j++)
      {
        if(arr[i] > arr[j])
```

```
        {
          temp = arr[i];
          arr[i] = arr[j];
          arr[j] = temp;
        }
      }
    }
    printf("\n");
    //Displaying elements of array after sorting
    printf("Elements of array sorted in ascending order: \n");
    for (i=1; i<=5; i++)
    {
      printf("%d ", arr[i]);
    }
    return 0;
}
```

Output:



Problem No. 06

Name of the problem:

Write a program in C program to Insert an Element in an Array.

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Insert an Element in an Array
#include<stdio.h>
int main()
{
   char a[0],i;
   printf("Enter an element\n");
   scanf("%c",&a[0]);
   printf("Entered element is %c",a[0]);
   return 0;



}
```

Output:

```
"E:\Programming\my files\C++\array\array_6.exe"
Enter an element
1
Entered element is 1
Process returned 0 (0x0)   execution time : 4.
253 s
Press any key to continue.
```

Problem No. 07

Name of the problem:

Write a program in C to print all unique elements in an array

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Print all unique elements of an array
#include <stdio.h>
int main()
{
   int a[100],n,ctr=0;
   int i,j,k;
     printf("\tPrint all unique elements of an array\n");
     printf("Input the number of elements to be stored in the array: ");
     scanf("%d",&n);
     printf("Input %d elements in the array :\n",n);
     for(i=0;i<n;i++)
      {
                printf("element - %d : ",i);
                scanf("%d",&a[i]);
      }
   printf("\nThe unique elements found in the array are:\n");
   for(i=0; i<n; i++)
```

```
  {
     ctr=0;
     for(j=0,k=n; j<k+1; j++)
     {
        /*Increment the counter when the search value is duplicate.*/
        if (i!=j)
        {
                        if(a[i]==a[j])
         {
            ctr++;
          }
         }
     }
    if(ctr==0)
     {
       printf("%d ",a[i]);
     }
  }
    printf("\n\n");
  return 0;
}
```

Output:

```
"E:\Programming\my files\C++\array\array_7.exe"
        Print all unique elements of an array

Input the number of elements to be stored in
the array: 6
Input 6 elements in the array :
element - 0 : 1
element - 1 : 1
element - 2 : 1
element - 3 : 3
element - 4 : 4
element - 5 : 5

The unique elements found in the array are:
3 4 5
```

Lab Report 10

Problem No. 01

Name of the problem:

Write a program in C to find the square of any number using the function

Introduction

Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//determine square value by user define function
#include <stdio.h>
int main()
{
   int a, result;
   printf("Going to calculate the square value\n");
   printf("Input any number of square:");
   scanf("%d",&a);
   result=sum(a);
   printf("The square of %d is:%d",a,result);
}

int sum(int a)
{
   return (a*a);}
```

Output:

```
"E:\Programming\my files\C++\Fuction\user_diff_func 1.exe"
Going to calculate the square value
Input any number of square:12
The square of 12 is:144
Process returned 0 (0x0)   execution time : 1.
217 s
Press any key to continue.
```

Problem No. 02

Name of the problem:

Write a program in C to swap two numbers using a function

Introduction

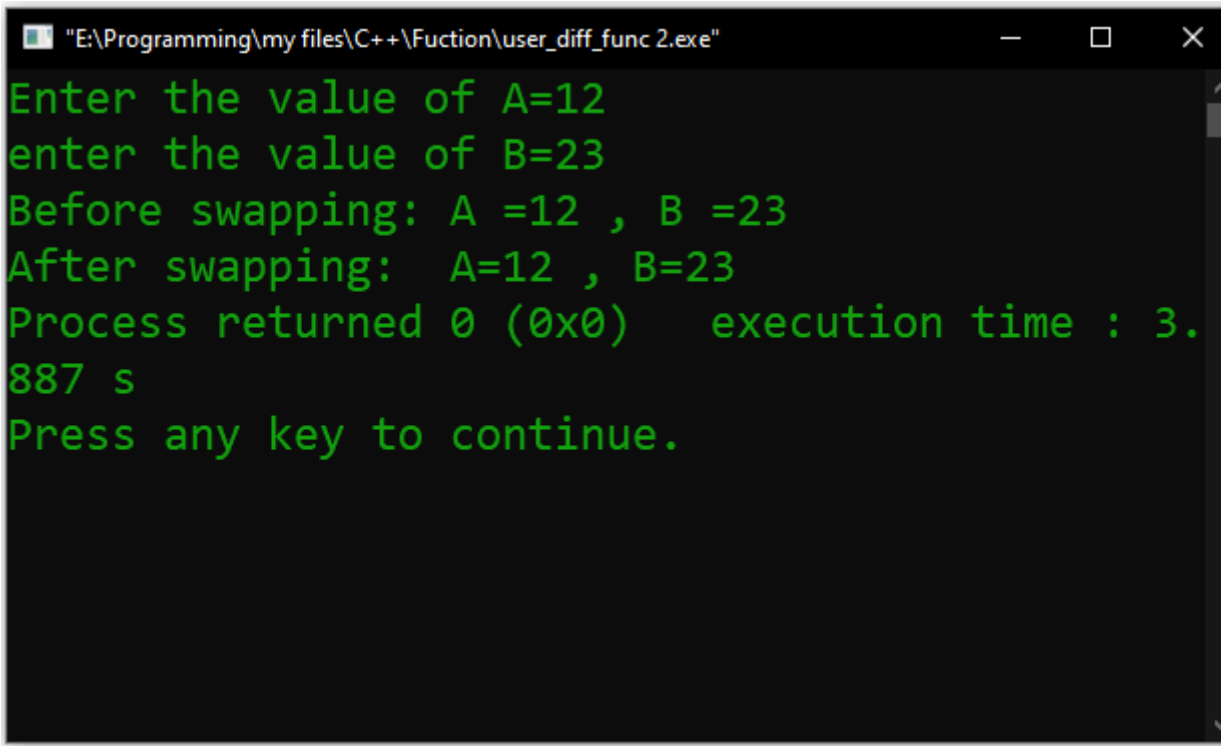
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//swiping value by user define function
#include <stdio.h>
int main()
{
   int a,b,c,d,e;
   printf("Enter the value of A=");
   scanf("%d",&a);
   printf("enter the value of B=");
   scanf("%d",&b);
   printf("Before swapping: A =%d , B =%d\n",a,b);

   swipe();
   d=swipe(a);
   e=swipe(b);

   printf("After swapping:  A=%d , B=%d", d,e);
```

```
}

int swipe(int a,int b,int c)
{
   c=a;
   a=b;
   b=c;

   return(a,b);
}
```

Output:



Problem No. 03

Name of the problem:

Write a program in C to check a given number is even or odd using the function

Introduction

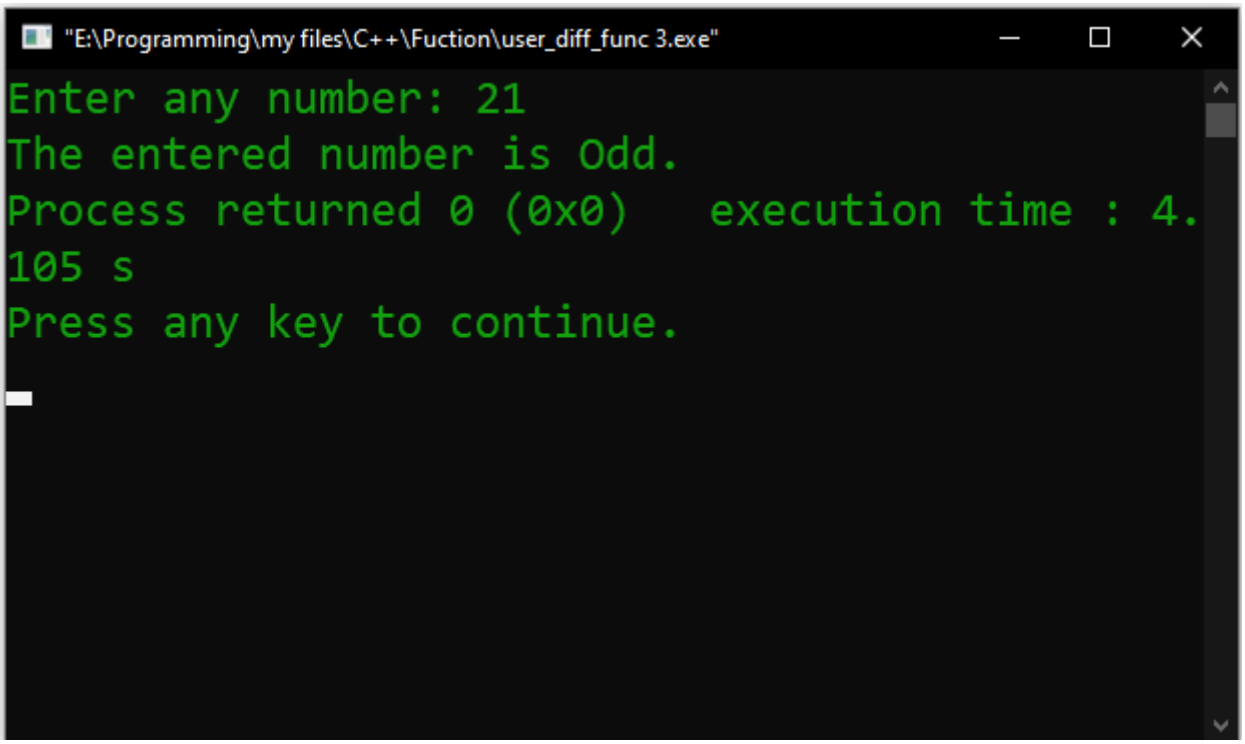
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//Even Odd (user define function)
#include <stdio.h>
int main()
{
   int a;
   printf("Enter any number: ");
   printf("");
   scanf("%d",&a);

   if(a%2==0)
      even();
   else
      odd();
}
even(int a)
{
   printf("The entered number is Even.");
```

```
}
odd(int a)
{
   printf("The entered number is Odd.");
}
```

Output:



Problem No. 04

Name of the problem:

Write a program in C to get the largest element of an array using the function.

Introduction

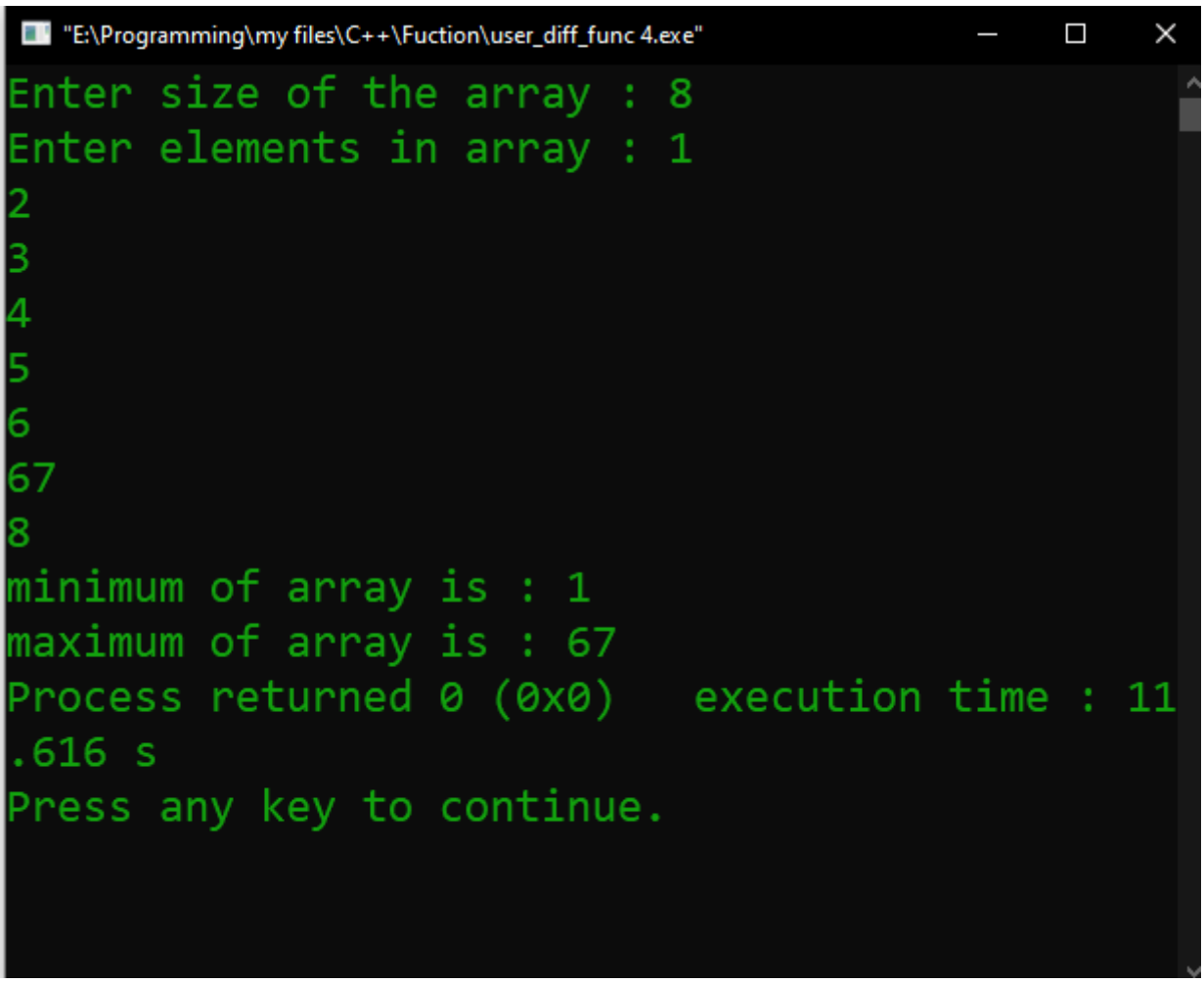
Implementing program in C is very interesting. To implement the problem given above we should remember the procedure to display outcome. That is, we should consider about the required inputs, the particular process that will be applied upon the input data and the output whether it is desired or not. The given problem gives me an opportunity to have an exercise on the procedure.

Program:

```
//find Maximum value from array numbers user define function
#include <stdio.h>
 int main()
{
   int a[1000],i,n,sum;
   printf("Enter size of the array : ");
   scanf("%d", &n);
   printf("Enter elements in array : ");
   for(i=0; i<n; i++)
   {
      scanf("%d",&a[i]);
   }
   sumofarray(a,n);
}
int sumofarray(int a[],int n)
 {
                int min,max,i;
```

```
            min=max=a[0];
  for(i=1; i<n; i++)
  {
     if(min>a[i])
                min=a[i];
                 if(max<a[i])
                  max=a[i];
  }
 printf("minimum of array is : %d",min);
  printf("\nmaximum of array is : %d",max);
 }
```

Output:



THE END